Paired movers

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Abstract

Consider two workers who both move from firm A to firm B. If firm and person effects are additively separable, then the pay gap between them at A should perfectly predict the gap at B, so paired movers are the simplest feature of the data that reveals match effects. Three instruments for this gap—its own lag, predetermined characteristics, and the pair's gap at a third firm—test three related hypotheses: additive separability (no match effects), exogenous mobility (no mobility on match effects), and exogenous mobility jointly with uncorrelated match effects. In Brazilian and Italian data, we reject all three hypotheses. A job search model estimated on these moments implies that match effects are about half as variable as firm effects and that counterfactuals based on estimated firm effects are wrong even in sign. In simulated data, standard specification tests supporting additive separability and exogenous mobility are biased or underpowered, reconciling our results with the literature.

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A large literature is based on the Abowd, Kramarz, and Margolis (1999) (AKM) wage equation where the (log) wage can be written as the sum of a person effect, a firm effect, and an independent error term (e.g., Card, Heining, and Kline (2013), Card, Cardoso, and Kline (2016), Sorkin (2018), and Gerard et al. (2021)). A causal interpretation of the firm effect is justified either by the assumption of additive separability (there are no match effects) or of exogenous mobility (workers do not move on the basis of match effects).

This equation is controversial because it conflicts with many canonical models of wage determination and mobility (see, e.g., Gibbons et al. (2005), Eeckhout and Kircher (2011), and Borovičková and Shimer (2024)). Nonetheless, three pieces of evidence are supportive of this specification. First, Card, Heining, and Kline (2013) introduce what we call the event study test for exogenous mobility, and show that for movers across firms the change in the firm effects does a good job of predicting changes in earnings. Second, Card, Heining, and Kline (2013) look at the change in adjusted-R^2 from an additively separable model (AKM) to a non-additively separable model and find that the change is small, suggesting a minimal role for match effects. Finally, Bonhomme, Lamadon, and Manresa (2019) develop a way of estimating interactions in the wage equation and find minimal deviations from additive separability.

In this paper, we revisit this evidence by exploring a novel feature of the data: *paired movers*, two workers who both move from firm A to firm B. Paired movers are the simplest feature of the data that lets us see match effects. Match effects are an interaction between a worker and a firm, so seeing them takes two workers at the same two firms. The earnings gap between the two workers at a firm removes the firm effect, and the difference in that gap across the two firms removes the person effects, which leaves match effects and transitory error. We show that different instruments for the earnings gap allow us to test different hypotheses about match effects: first, whether they exist (additive separability), and, second, whether worker mobility conditions on them (exogenous mobility). We also test a third and stronger hypothesis—exogenous mobility combined with uncorrelated match effects—which avoids taking a stance on the source of match effects. We find evidence against all three hypotheses. We estimate a model of endogenous mobility and use it to show why this evidence matters for applied questions. Finally, we use a combination of the model and the paired movers to revisit each of the three pieces of evidence we described in the last paragraph. In each case, we find that it is less compelling than previously thought.

We begin by explaining how additive separability is stronger than exogenous mobility. Suppose that there are match effects in the wage equation but there is random assignment of workers to firms. In this case, we can consistently estimate the average treatment effects of firms because the match effects average out. So exogenous mobility is satisfied. But additive separability is not satisfied because there are match effects. Thus, this distinction motivates developing tests for the presence of match effects and for whether mobility depends on match effects.

We then introduce the paired movers and use them to test additive separability. We regress the destination gap on the origin gap and call the slope the paired-mover coefficient. Under additive separability the origin gap perfectly forecasts the destination gap so the coefficient is 1. Transitory

error in the origin gap would bias the slope toward zero, so we instrument with the lagged origin gap, which captures every source of match effects at the origin firm.

A coefficient below 1 is also informative about how large match effects are. Under exogenous mobility, one minus the paired-mover coefficient is the share of the persistent variance of the gap that comes from match effects, and the autocovariance of the origin gap measures that persistent variance. Combining the two gives an estimator for the variance of match effects. Because it uses only covariances across periods, the estimator is robust to heteroskedasticity in the transitory error.

We then test exogenous mobility directly. We run the same regression but instrument the origin gap with a predetermined characteristic: the gender composition of the pair. When the gender pay gap differs across firms, that difference is a match effect. Unlike the lagged gap, gender contains no information specific to the origin firm, so under exogenous mobility it predicts the gap equally well at the origin and the destination, and the coefficient is 1 even when match effects exist. The coefficient departs from 1 only if gender pay gaps differ systematically between the firms workers leave and the firms they join, which means workers move on this match effect. The test therefore detects only endogenous mobility on gender-specific firm effects.

Our third test does not tie match effects to an observable characteristic. It uses paired double movers: paired movers who move together a second time, so we observe their gap at three firms. We run the same regression but instrument the origin gap with the gap at the third firm. The coefficient is then a ratio of two cross-firm covariances of the gap, so it is 1 when the gap covaries equally across every pair of firms. This equality is the testable implication of exogenous mobility combined with uncorrelated match effects, meaning that a worker's match effects are uncorrelated across firms and a firm's are uncorrelated across workers. We test the two assumptions jointly because they tend to fail together: when workers choose jobs based on match effects, the match effects of the jobs they hold become correlated.

We use administrative data from two countries: Italy and Brazil. For the Italian data, we use the Veneto Worker History data. For the Brazilian data, we use the RAIS data. Because the Brazilian dataset is much larger (and less noisy), we focus on the results from Brazil.

The paired movers are a non-trivial share of the data: 8% of mover-years and 15% of movers. The most important difference between paired movers and all movers is that paired movers start from—and end up at—much larger firms. If paired movers arise mostly through chance, then this pattern is what we expect: large firms send many workers to the same large destinations, so two workers often make the same move. We show that this mechanism explains most paired mobility.

We document substantial deviations from additive separability. As a benchmark, the paired-mover coefficient is 0.97 among stayers at the same firm, while among firm switchers it is 0.76. Under the assumption of exogenous mobility, our estimator finds that the variance of match effects is about 40% of the Kline, Saggio, and Sølvsten (2020) (KSS)-corrected variance of firm effects in the full sample.

We then implement our tests for exogenous mobility. For predetermined characteristics, we consider gender, age, and education. In all cases, we can decisively reject exogenous mobility,

with especially large rejections for age and gender. In the joint test of exogenous mobility and uncorrelated match effects using paired double movers, we again reject.

To develop an estimate of the variance of match effects taking into account endogenous mobility, we estimate a random job search model where workers are heterogeneous in productivity and move on the basis of both a firm effect and a match effect. Quantitatively, the model finds that the variance of match effects in the steady state distribution is about 25% bigger than that found by our exogenous mobility estimator, or over half of the variance of firm effects in the full sample.

These findings matter when AKM is used to run reallocation counterfactuals. AKM is used for two purposes: to decompose the variance of earnings (as in Card, Heining, and Kline (2013)), and to run counterfactuals that move workers across firms (as in the sorting component of the gender gap in Card, Cardoso, and Kline (2016)). In our model, match effects and endogenous mobility barely affect the first: the AKM estimator of the variance of firm effects is biased *down* by less than 5% relative to the truth. They do, however, generate large biases in the second type of exercise. Consider a prototypical reallocation exercise: rank firms on the basis of their estimated firm effects and move workers from the bottom 20% of firms to the top 20%. In the model, we can do this for workers who received—and rejected—such an offer, so we know their counterfactual earnings. The change in firm effects predicts earnings increases of 27 log points. Instead, their actual counterfactual earnings decrease by 17 log points. The reallocation exercise is wrong in both sign and magnitude.

Turning to the first piece of evidence supporting the AKM specification, this model generates an event study coefficient of 1.00. Thus, the data from the model passes the event study test, which is surprising given that it features endogenous mobility. Splitting the data, we find a coefficient of 0.54 on employer-to-employer (EE) transitions (about 40% of the data) and 1.04 on employer-to-unemployment-to-employer (EUE) transitions. The pooled data passes the event study test because the EE moves see bigger increases in firm effects and in earnings than the EUE moves, which is a generic feature of endogenous mobility. Because the two samples have different means, pooling them produces a coefficient of 1—a mechanism analogous to Simpson's paradox. This between-sample pooling mechanism offers a new explanation for why the event study can be low-powered.

Turning to the second piece of evidence, we document two sources of downward bias in the comparison of the adjusted-R^2 from AKM and an interacted model: endogenous mobility and short panels. With endogenous mobility and short panels, this estimator is downward biased by almost an order of magnitude. Thus, the small variance of match effects found by this estimator is consistent with a much larger role for match effects than previously thought.

Turning to the third piece of evidence, we implement the static Bonhomme, Lamadon, and Manresa (2019) (BLM) estimator in our data. Typically, the BLM estimator finds some deviations from additive separability and so we want to translate its deviations into our metrics. We simulate data from the BLM model estimated on our data and then run the paired-mover test in the simulated data. The BLM estimator is unable to accurately recover the paired-mover coefficient. Monte Carlo evidence shows that this failure is because BLM struggles to reliably estimate mobility

patterns. We conclude that our test points to a new feature of the data.

Literature: The most closely related paper is Kantenga and Law (2016). Their Section 6 develops a statistical test for additive separability based on the same feature of the data that this paper exploits: two workers observed at the same two firms (albeit not restricted to the same years) should have constant earnings gaps up to concerns about transitory error. We develop a more transparent approach to looking at this implication. We also use paired movers to develop an estimator for the variance of match effects and to test for exogenous mobility.

Woodcock (2015) develops estimators of match effects in the context of the AKM model. One is the orthogonal match effects estimator which defines the match effect as the mean residual of the AKM model within worker-firm match. As Kline (2024) emphasizes, the estimation of many parameters in the AKM model relies on indirect contrasts where it is less transparent what feature of the data is driving results. The present paper builds from the minimal feature of the data that identifies match effects.

Kline (2024) uses Veneto data and its Table 1 looks at the origin-destination firm pairs (edges) where there are two or more movers, which is the same feature of the data that this paper relies on. It shows that the AKM model captures 84% of the variance of edge effects generated by looking directly at the movers. This paper looks at a different feature of these edges: whether the earnings gaps between workers are stable with the moves. In this sense, the exercise in this paper is reminiscent of a pre-processing step in Hagedorn, Law, and Manovskii (2017) where they find that the ranking of workers across firms is not stable and they develop an approach to average this out (see their section 4.2).

A number of "AKM" papers present evidence that is consistent with match effects. Most directly, Beauregard et al. (2026) allow the person effect to depend on whether they are matched to a union or non-union firm. A number of papers ask whether the change in the firm effects can explain the change in earnings for a select subset of workers. Sometimes firm effects do a poor job of explaining these changes. For example, Lachowska, Mas, and Woodbury (2020) find that the change in firm effects does not on average explain the earnings losses of displaced workers. Relative to this literature which relies on having a specific shock to mobility or a particular model of match effects, this paper presents an omnibus approach to testing for and quantifying the role of match effects.

Conceptually, this paper is related to Borovičková and Shimer (2024). They develop a model with endogenous mobility because of selective hiring. The paper develops analytical and numerical evidence that the model passes the event study test of Card, Heining, and Kline (2013), despite an economic structure that violates exogenous mobility. We show that the reason their model passes the event study test is the between-sample effect that we emphasized above. More broadly, while the point of Borovičková and Shimer (2024) is to show analytically that it is possible for an economic structure featuring match effects to deliver additively separable wage equations, our goal is to directly test this functional form restriction.

1 Paired movers

In this section, we introduce paired movers. We first write down the wage equation with match effects and formally state the assumptions of exogenous mobility and additive separability. We show how to use paired movers to test for—and quantify the presence of—match effects. We do so under the assumption of exogenous mobility. We then use paired movers to test implications of exogenous mobility using predetermined characteristics as instruments. Finally, we back away from a parametric model of match effects (i.e., match effects that only depend on observables) and use paired double movers to develop a test for the joint hypothesis of exogenous mobility and uncorrelated match effects.

1.1 Set-up

Wage equation, exogenous mobility, and additive separability: We write the log earnings of worker i at firm $j = j(i,t)$ in period t as

$$y_{it} = \alpha_i + \psi_{j(i,t)} + \beta X_{it} + \mu_{ij(i,t)} + \tilde{\epsilon}_{ij(i,t)t}, \tag{1}$$

where α_i is a person effect, ψ_j is a firm effect, and X_{it} are time-varying covariates. The error term is split into a permanent component, $\mu_{ij} \equiv \mathbb{E}[\epsilon_{ijt}|i,j]$, and a transitory component, $\tilde{\epsilon}_{ijt} \equiv \epsilon_{ijt} - \mu_{ij}$. Here, ϵ_{ijt} is the potential error of worker i at any firm j in period t, so μ_{ij} is defined over all draws, not only in the periods in which i and j match.[1] Setting $\mu_{ij} = 0$ gives the standard AKM wage equation. We call the μ_{ij} match effects; they are statistical and not economic objects. We treat $(\alpha, \psi, X, \mu, \tilde{\epsilon})$ as random draws from a population distribution, and let $\mathcal{G} \equiv \{j(i,t)\}$ collect all assignments of workers to firms.

The AKM literature uses one of two assumptions to justify a causal interpretation of ψ_j.

Assumption 1 (Exogenous mobility)**.** *Assume that:*

$$\mathbb{E}[\tilde{\epsilon}_{ijt}|\mathcal{G}, \alpha, \psi, X, \mu] = 0, \forall i, j, t,$$

and

$$\mathbb{E}[\mu_{ij}|\mathcal{G}, \alpha, \psi, X] = 0, \forall i, j.$$

Assumption 2 (Additive separability)**.** *Assume that:*

$$\mathbb{E}[\tilde{\epsilon}_{ijt}|\mathcal{G}, \alpha, \psi, X] = 0, \forall i, j, t,$$

and

$$\mu_{ij} = 0 \ \forall \ i, j.$$

[1] This definition parallels the way firm and person effects are defined, and equation (3) in Woodcock (2015).

Exogenous mobility lets workers sort on α and ψ (and on non-wage factors), but not on the match effect or the transitory error, so match effects may exist but average out. Additive separability says there are no match effects. Hence, additive separability implies exogenous mobility. For example, with match effects and random assignment, Assumption 1 holds, but not Assumption 2.

Transitory error: As we discuss in more detail below, separating match effects from the transitory error relies on parametric assumptions about that error. Here we introduce our assumption.

Assumption 3 (Uncorrelated transitory errors)**.** *We split this assumption into two parts:*

(i) (Serially uncorrelated within a match). Within a match, the transitory error is serially uncorrelated:

$$\mathbb{E}[\tilde{\epsilon}_{ijt}\tilde{\epsilon}_{ij,t-\ell}|\mathcal{G},\alpha,\psi,X,\mu]=0,\ \textit{for all}\ \ell\geq 1.$$

(ii) (Uncorrelated across matches). For any two distinct worker-firm matches $(i,j)\neq(i',j')$ and any periods $t\neq s$,

$$\mathbb{E}[\tilde{\epsilon}_{ijt}\tilde{\epsilon}_{i',j',s}|\mathcal{G},\alpha,\psi,X,\mu]=0.$$

Part (i) separates the permanent match effect from transitory error. This assumption imposes no restrictions on the levels of the transitory variances, $\text{Var}(\tilde{\epsilon}_{ijt})$, which may vary arbitrarily across workers and over time. The estimator we present below is thus robust to such heteroskedasticity.

We can replace Assumption 3(i) with a weaker assumption on the error term: that the transitory error follows an MA$(k-1)$ process within a match so that $\mathbb{E}[\tilde{\epsilon}_{ijt}\tilde{\epsilon}_{ij,t-\ell}|\mathcal{G},\alpha,\psi,X,\mu]=0,$ for all $\ell\geq k$. The results we derive below are for $k=1$. The extensions to $k>1$ are identical except that they require the use of deeper lags. In our empirical section, we also use these deeper lags.

Assumption 3(ii) rules out three forms of correlation in the transitory error: first, between the same worker at different firms $(i=i',j\neq j')$; second, between different workers at the same firm $(i\neq i',j=j')$; and, third, between different workers at different firms $(i\neq i',j\neq j')$. Ruling out correlation across matches makes the transitory error pure noise.

The paired-mover population: The estimator is defined on the population of *paired movers*: pairs of workers who are observed at the same origin firm and subsequently move together to the same destination firm.

Definition 1. *A pair of workers (i,i') is a* paired mover *if there exist firms $A\neq B$ and periods t and t' such that both workers are employed at A in period t and B in period t':*

$$j(i,t)=j(i',t)=A\ \textit{and}\ j(i,t')=j(i',t')=B$$

for $t<t'$. We require that each worker is observed at the origin firm for at least two years, which allows us to instrument. Denote by $\mathcal{P}$ the set of such pairs. Expectations and variances written $\mathbb{E}[\cdot|\mathcal{P}]$ and $\text{Var}(\cdot|\mathcal{P})$ are taken over this population.

Within a pair, we order the two workers at random so that there is no systematic relationship between which worker is labeled i and the sign of the earnings gap. This random ordering has consequences for demeaning that we discuss below.

The gap between paired movers: The feature of the data we are interested in is the earnings gap between the paired movers. For two workers $(i, i') \in \mathcal{P}$ employed at a common firm j in period t define the earnings gap (where we suppress covariates for simplicity; in the empirical section, we work with residualized earnings)

$$Z^j_{ii't} \equiv y_{it} - y_{i't} = (\alpha_i - \alpha_{i'}) + (\mu_{ij} - \mu_{i'j}) + (\tilde{\epsilon}_{ijt} - \tilde{\epsilon}_{i'jt}),$$

where the firm effect differences out. It is convenient to have notation for the differences in each component:

$$\Delta\alpha \equiv \alpha_i - \alpha_{i'}, \qquad \Delta\mu^j \equiv \mu_{ij} - \mu_{i'j}, \qquad \nu^j_t \equiv \tilde{\epsilon}_{ijt} - \tilde{\epsilon}_{i'jt},$$

so that $Z^j_t = \Delta\alpha + \Delta\mu^j + \nu^j_t$, where we suppress the pair subscripts for simplicity.

1.2 The paired-mover coefficient under additive separability

Consider a pair that moves from an origin firm A to a destination firm B. Under additive separability (and ignoring transitory error) the earnings gap is the difference in the person effects and so is stable across firms. Thus, a lack of stability suggests deviations from additive separability. Formally, we consider the following linear model:

$$Z^B_{ii't'} = \beta_0 + \beta_1 Z^A_{ii't} + \varepsilon_{ii't}. \tag{2}$$

Under additive separability and without transitory error, both gaps equal the gap in person effects, so OLS gives $\beta_1 = 1$. Of course, the regressor, $Z^A_{ii't}$, contains the origin noise, $\nu^A_{ii't}$, so OLS is biased towards zero. To address this issue, we instrument $Z^A_{ii't}$ with the gap between the same two workers at the origin firm one period earlier, $Z^A_{ii',t-1}$. Under Assumption 3(i), the origin noise is serially uncorrelated and so the lagged gap is a valid and relevant instrument.

We collect this discussion in a formal result.

Result 1. *Consider estimating equation (2) by instrumental variables, where $Z^A_{ii't}$ is instrumented with $Z^A_{ii't-1}$. Suppose Assumption 2 (additive separability), Assumption 3 (uncorrelated transitory errors), and the relevance condition that* $\mathrm{Var}(\alpha_i - \alpha_{i'}|\mathcal{P}) > 0$ *hold. Then*

$$\operatorname{plim} \hat{\beta}_1 = 1.$$

Proof. This proof and all subsequent proofs can be found in Appendix A. □

We refer to β_1 as the paired-mover coefficient. This result says that under additive separability it equals 1. Thus, deviations from 1 are indicative of the presence of match effects.

Instrumenting with the lagged gap at the origin firm is important for testing the hypothesis of additive separability. As we show below, other natural sets of instruments—predetermined characteristics such as gender, or earnings gaps in a third spell—test different hypotheses than additive separability.

The cost of this instrument is that we restrict the within-match transitory error, which is stronger than is standard in the literature (e.g., Kline, Saggio, and Sølvsten (2020)). We need to make some restriction to separate the persistent and transitory parts of the gap. The concern with our restriction is that if a component of pay lasted two years but was not permanent, we would attribute it to the match effect. In practice, however, this attribution is reasonable. Median tenure is around two years, both among all workers and among paired movers at their origin firm (Table 1). So a component of pay that persists for two years lasts as long as a typical job. Over the life of that job, it is effectively permanent, which is what a match effect is.

1.3 An estimator for the variance of match effects under exogenous mobility

In the last subsection, we showed how to test the null of no match effects. Here, we quantify the importance of match effects by developing an estimator for the variance of match effects.

1.3.1 Set-up

Target parameter: The target parameter is the variance of match effects in the paired-mover population, which is an omnibus notion of match effects that captures all possible sources. Under exogenous mobility the mean of match effects within each firm is zero, but for notational consistency it is helpful to work with the demeaned version. Let $\bar{\mu}_j \equiv \mathbb{E}[\mu_{ij}|j, \mathcal{P}]$ be the mean match effect at firm j among the paired movers so that $\tilde{\mu}_{ij} \equiv \mu_{ij} - \bar{\mu}_j$ is the within-firm demeaned match effect. The target parameter is thus $\text{Var}(\tilde{\mu}_{ij}|\mathcal{P})$.

Uncorrelated match effects: Paired movers identify match effects only through the within-pair earnings gap, which leaves two parts of match effects unidentified. First, any part of a worker's match effect that persists from firm A to firm B enters the gap identically at both firms, and so cannot be separated from the person effect. Second, any part of the match effect that the two workers share at a firm cancels out along with the firm effect. We therefore need to rule out correlation in match effects both for the same worker at different firms and for different workers at the same firm. We make the following assumption:

Assumption 4 (Uncorrelated match effects across matches)**.** *For any two distinct worker-firm matches* $(i, j) \neq (i', j')$*,*

$$\mathbb{E}[\mu_{ij}\mu_{i',j'}|\mathcal{G}, \alpha, \psi, X] = 0.$$

In words, match effects are uncorrelated across distinct matches: for the same worker at different firms, for different workers at the same firm, and for different workers at different firms. Assumption 4 is the analogue for match effects of Assumption 3(ii) (uncorrelated transitory error across matches).

This assumption is distinct from exogenous mobility, which restricts the conditional mean of match effects but not their second moments. Match effects among realized matches can be correlated for two reasons. First, the potential match effects can be correlated. Suppose that match effects have a worker-sector component. Then the μ_{ij} are correlated within a worker across firms in the same sector. This correlation by itself does not violate exogenous mobility, since it is a property of match effects and not mobility patterns. Second, selection can create correlation. In the model in Section 4, match effects are independent draws, but workers choose jobs on the basis of them (violating Assumption 1 (exogenous mobility)), and the match effects of the jobs they hold are correlated as a result. The economic content of exogenous mobility is that workers do not choose firms on the basis of match effects, which rules out the second source. Together, the two assumptions imply that match effects are uncorrelated among realized matches, which the estimator below assumes and which we test in Section 1.5.

1.3.2 The estimator

We first introduce two moments that we use to form our estimator for the target parameter. Later on in the paper, we will target these two moments since they are the features of the data most directly informative about match effects. We then show how Assumptions 1, 3, and 4 let us identify the variance of match effects.

Moment one: the instrumental variables regression: We consider the same IV regression as in Result 1. We now analyze its probability limit under a different set of assumptions.

Result 2. *Consider an instrumental variables version of equation (2) where $Z^A_{ii't}$ is instrumented with $Z^A_{ii't-1}$. Suppose Assumptions 1, 3, and 4, the fact that workers within a pair are ordered randomly, and the relevance condition that* $\mathrm{Var}(\alpha_i - \alpha_{i'}|\mathcal{P}) > 0$ *hold. Then*

$$\operatorname{plim} \hat{\beta}_1 = \frac{\mathrm{Var}(\Delta\alpha \,|\, \mathcal{P})}{\mathrm{Var}(\Delta\alpha \,|\, \mathcal{P}) + 2\,\mathrm{Var}(\tilde{\mu}_{ij} \,|\, \mathcal{P})}.$$

This result is a more general version of Result 1: if $\mathrm{Var}(\tilde{\mu}_{ij}|\mathcal{P}) = 0$, then $\operatorname{plim} \hat{\beta}_1 = 1$, and a coefficient below 1 is indicative of the presence of match effects.

The heuristic intuition of this IV regression is that deviations from 1 are indicative of deviations from additive separability. Why does this result imply that the slope should be below 1, rather than potentially above 1? The IV estimand is $\mathrm{Cov}(Z^B, Z^A_{-1})/\,\mathrm{Cov}(Z^A, Z^A_{-1})$. The lagged origin gap co-moves more closely with the origin gap than with the destination gap, so the ratio is below 1.

Moment two: the autocovariance of the gap: The second moment is the autocovariance of the origin gap across adjacent periods. Because the signal is constant within the spell and the noise is serially uncorrelated, the covariance of the gap across periods captures the signal variance.

Result 3. *Under Assumptions 1, 3, and 4 and the fact that workers are ordered randomly within pairs*

$$\operatorname{plim} \widehat{\operatorname{Cov}}(Z_t^A, Z_{t-1}^A) = \operatorname{Var}(\Delta\alpha \mid \mathcal{P}) + 2\operatorname{Var}(\tilde{\mu}_{ij} \mid \mathcal{P}).$$

An estimator for the variance of match effects: The two moments identify the target parameter. The autocovariance tells us the variance of the persistent part of the gap, while one minus the IV coefficient, $1-\hat{\beta}_1$, tells us the share of this variance due to match effects. Combining,

$$\operatorname{Var}(\tilde{\mu}_{ij}|\mathcal{P}) = \frac{1}{2}(1 - \operatorname{plim}\hat{\beta}_1)\operatorname{plim}\widehat{\operatorname{Cov}}(Z_t^A, Z_{t-1}^A).$$

The corresponding estimator replaces the population moments with their sample analogues:

$$\widehat{\operatorname{Var}}(\tilde{\mu}_{ij}|\mathcal{P}) = \frac{1}{2}(1 - \hat{\beta}_1)\widehat{\operatorname{Cov}}(Z_t^A, Z_{t-1}^A). \tag{3}$$

1.4 A direct test of exogenous mobility using predetermined characteristics

So far we have used paired movers to develop a test for additive separability and to quantify the variance of match effects under the assumption of exogenous mobility. We now use paired movers to develop a direct test for exogenous mobility using a predetermined characteristic as an instrument.

Let $f_i \in \{0, 1\}$ be an indicator for whether worker i is female. Because gender is time-invariant, in equation (1) it is absorbed into the person effect. We decompose the person effect into a gender component and a residual component, $\alpha_i = \bar{\gamma} f_i + \alpha_i^r$, where $\bar{\gamma}$ is the average gender gap and α_i^r is the residual person effect. We allow the gender gap to differ across firms so the wage equation (1) becomes

$$y_{it} = \alpha_i^r + \psi_{j(i,t)} + \gamma_{j(i,t)} f_i + \mu_{ij(i,t)} + \tilde{\epsilon}_{ij(i,t)t}, \tag{4}$$

where γ_j is the firm-specific gender gap (a female penalty when $\gamma_j < 0$).

Now suppose that we instrument for the earnings gap using the gender configuration of the pair. For a pair (i, i'), let

$$c_{ii'} \in \mathcal{C} \equiv \{MM, FF, MF, FM\}$$

record the genders of the two workers, and let $D_{ii'}^c \equiv \mathbf{1}\{c_{ii'} = c\}$. The instrument set is the gender-configuration dummies,

$$D_{ii'} \equiv (D_{ii'}^{FF}, D_{ii'}^{MF}, D_{ii'}^{FM})',$$

where the male-male configuration is omitted because we include an intercept. We estimate equation (2) by two-stage least squares with $D_{ii'}$ as the instruments. We have the following result, which we state informally.

Result 4. *(Informal). Consider estimating equation (2) by two-stage least squares, where $Z^A_{ii't}$ is instrumented with the gender-configuration dummies $D_{ii'}$, in the model with firm-specific gender gaps (equation (4)). Suppose a relevance condition holds, the natural extension of Assumption 1 (exogenous mobility) holds and at least one worker is female. Then:*

$$\text{Cov}(Z^B, D_{ii'}|\mathcal{P}) = \text{Cov}(Z^A, D_{ii'}|\mathcal{P}), \qquad \textit{and hence} \qquad \text{plim}\,\hat{\beta}_1^{gender} = 1.$$

In Appendix B (and Result A1) we develop a more formal version of this result which requires restating Assumption 1 in this new notation.

The gender instrument tests exogenous mobility because it is symmetric between the origin and destination firm: it contains no information specific to either, so each gender-configuration dummy has the same covariance with the earnings gap at the origin and destination firm. Under exogenous mobility, the gender-predicted gap can differ between A and B for any particular pair, but these differences average out. So the coefficient is 1 even when match effects exist. The coefficient departs from 1 only if mobility is systematically related to the firm-specific gender gap. For example, Card, Cardoso, and Kline (2016) document that higher-paying firms have bigger gender pay gaps. Because the firm-specific gender gap is a component of the match effect, if opposite-gender pairs tended to move to higher-paying firms, then this pattern would reflect mobility that conditioned on match effects, which would violate exogenous mobility.

The lagged-gap instrument tests a different hypothesis because it is asymmetric: it is origin-specific, and so more closely related to the gap at the origin firm (the denominator of the IV estimand) than to the gap at the destination firm (the numerator). Whenever match effects exist, the coefficient is therefore below 1, even if mobility is exogenous. Hence the asymmetric instrument tests additive separability, while the symmetric instrument tests exogenous mobility.

In the empirical section, we implement this test using gender, and also education and age. Appendix B shows the same logic applies to other predetermined characteristics.

1.5 A joint test for exogenous mobility and uncorrelated match effects

While the previous section developed a direct test of exogenous mobility, it only captures deviations from exogenous mobility that are specific to the characteristics used (i.e., the match effects are parameterized through observables). As we show below, gender-specific match effects account for less than 10% of the estimated match effects under exogenous mobility. Here we show how to use information in other spells to develop a joint test for exogenous mobility and uncorrelated match effects that captures broader deviations, though at the cost of going to a more restrictive sample. As discussed above, economic models with selection break these assumptions together (as does the model in Section 4). To implement this test, we introduce *paired double movers*.

Definition 2. *A pair of workers* (i, i') *is a* paired double mover *if there exist distinct firms* A, B, C *and periods* $t < t' < t''$ *such that both workers are employed at* A *in period* t, *at* B *in period* t', *and at* C *in period* t''*:*

$$j(i,t) = j(i',t) = A, \quad j(i,t') = j(i',t') = B, \quad \text{and} \quad j(i,t'') = j(i',t'') = C,$$

with $A \neq B$, $B \neq C$, *and* $A \neq C$. *Denote by* $\mathcal{P}_2 \subseteq \mathcal{P}$ *the set of such pairs. Expectations and variances written* $\mathbb{E}[\cdot \mid \mathcal{P}_2]$ *and* $\mathrm{Var}(\cdot \mid \mathcal{P}_2)$ *are taken over this population.*

For the test, we use Z^C as the instrument in equation (2).

Result 5. *Consider estimating equation (2) by instrumental variables where* $Z^A_{ii'}$ *is instrumented with* $Z^C_{ii'}$. *Suppose that Assumption 1 (exogenous mobility), Assumption 3(ii) (transitory errors uncorrelated across matches) and Assumption 4 (uncorrelated match effects) hold, and that* $\mathrm{Var}(\alpha_i - \alpha_{i'} | \mathcal{P}_2) > 0$. *Then*

$$\operatorname{plim} \hat{\beta}_1^{double} = \frac{\mathrm{Cov}(Z^B_{ii'}, Z^C_{ii'} | \mathcal{P}_2)}{\mathrm{Cov}(Z^A_{ii'}, Z^C_{ii'} | \mathcal{P}_2)} = 1.$$

Thus, deviations from 1 jointly test exogenous mobility and uncorrelated match effects. These two assumptions jointly say that the covariance of the paired-mover gap should be the same between any two spells. Deviations of the IV coefficient from 1 are indicative that it is not. We refer to the IV coefficient as the double-mover coefficient.

Implementation: The above test is under-specified in that there is researcher discretion about which firm gap is the outcome, which is the endogenous variable and which is the instrument. Each of the six assignments produces an IV estimand that is the ratio of two of the three cross-firm covariances. Switching the outcome and the endogenous variable just inverts the ratio, so there are three conceptually distinct estimands. We report the three specifications with $\hat{\beta}_1^{\text{double}} < 1$.

All three estimands equal 1 under exogenous mobility and uncorrelated match effects. Therefore, coefficient instability across the choice of estimands is a sign that the joint hypothesis of exogenous mobility and uncorrelated match effects fails.

2 Data

We use data from two countries: Brazil and Italy.

2.1 Brazilian data

We use *Relação Anual de Informações Sociais* (RAIS), which is a matched employer-employee dataset. Each observation is a job spell and an employer is an establishment within a firm. We have data from 2006 to 2022. We include only the worker's dominant job, which is the job with

the highest earnings in each year. We drop workers on temporary contracts, public servants, and those with zero earnings or missing gender or education information. We include workers aged 18 to 64. This sample is *all workers*.

We define a mover as a worker who changes employers between two consecutive years and separates from the previous employer. We exclude workers who change their dominant job but remain employed at the original firm. We also exclude workers who transfer between establishments within the same firm. This sample is *all movers*.

To construct the sample of *paired movers*, we make a few restrictions. A move is defined by origin firm, destination firm, and year. First, we focus on paired movers where the workers are present for two years in the origin firm. This restriction allows us to instrument the earnings at the origin firm. Second, we require that no more than 30% of the workers in the origin firm go to the destination firm, and that no more than 30% of employment in the destination firm came from the origin firm (Benedetto et al. (2007)). In many cases, there are more than two workers making a particular move. In this case, we randomly match workers and each worker only appears in one pair. If there is an odd number of workers making a particular move, then we throw out one worker. In addition, within the pair we randomly order workers so that sometimes the earnings gap is positive and sometimes negative. We do so because in Section 1 our derivations used the fact that workers were randomly ordered.

To construct the sample of *paired double movers*, we also make a few restrictions. A double move is defined by the origin, middle, and destination firms, and the years of the two moves. We maintain the same restrictions as for paired movers in terms of worker flows between firms. We focus on cases where the pair of workers were together in their last year at the origin firm and the first year at the destination firm.

We also construct a sample of *paired stayers*, where the origin and destination firm are the same. For computational reasons, we take a 5% random sample.

We construct four other samples to estimate AKM models. We start from the sample of *all workers* defined above. Due to the large size of the data, we take a 25% random sample of the workers who appear in 2015 and construct a panel of these workers from 2012 to 2017. Building on this first sample, the second sample includes only workers who change employers at least once between 2012 and 2017—that is, we drop the stayers. The third sample includes only workers in the second sample who are in the paired-mover sample (whose paired move is between 2012 and 2017). The fourth sample is the subset of the third sample who are paired double movers.

Earnings are the average log hourly wage in the year. Job contracts in Brazil are defined by monthly wages and weekly working hours. We compute hourly wages as the average monthly wage divided by an estimate of monthly hours (weekly hours $\times$ 30/7) following Gerard et al. (2021), and deflate values to 2019 BRL. Firm size is the number of workers employed on December 31 of each year.

Age is measured on December 31 of the year. Years of education is the current level of education completed using the *Instituto Brasileiro de Geografia e Estatística* (IBGE) classification. Tenure is

measured in months using the date of hire and the date of separation (or December 31). Occupation is classified using three-digit *Classificação Brasileira de Ocupações* (CBO) codes. Sectors are defined using two-digit *Classificação Nacional de Atividades Econômicas* (CNAE) codes.

For most of the analysis, we work with residualized earnings. We construct residualized earnings in two steps. First, we estimate the role of covariates using only variation in wages within a worker-firm match. Our baseline set of covariates is a third-degree polynomial in age, interacted with gender and four education categories (no primary education, primary education, high school, and some college or more). Following Card et al. (2018), we normalize age around 40 years old and omit the linear term. Second, we construct residualized earnings by subtracting off the estimated effect of covariates. We use residualized earnings almost everywhere. In the AKM analysis, we show how the conventional AKM regression compares to the results using residualized earnings.

2.2 Veneto data

We use cleaning code from Kline, Saggio, and Sølvsten (2020) and so our description follows their computational appendix. We use data from 1985 to 2001. We use workers aged 18 to 64. The earnings concept is daily wages. We use the worker's dominant (highest-earnings) job. We exclude workers with low wages (below 5 euros or zero days worked), large wage changes (more than doubling), employed in the public sector, having more than ten jobs, or missing gender. The industry codes are five-digit ATECO 91 codes. The employers in the Veneto data are firms. We construct paired movers identically to what we do in the Brazilian data.

3 Results

In this section, we present our tests for the presence of match effects (additive separability), mobility on the basis of match effects (exogenous mobility), and the joint hypothesis of exogenous mobility and uncorrelated match effects.

3.1 Additive separability

3.1.1 Descriptive statistics

The goal of this part of the paper is to learn about match effects in the population. We build on the firm effects literature, which extrapolates from the sample of all movers (in the connected set) to all workers. Table 1 shows summary statistics among all workers, all movers and paired movers. Paired movers are a sizeable portion of the data: about 8% of mover-years and 15% of mover-workers.

Movers are negatively selected relative to all workers: they have shorter tenure and lower earnings. The (large and influential) firm effects literature is therefore already extrapolating from a quite different population from all workers. Paired movers are more similar to all workers than all

movers are: on tenure and earnings they are either less negatively selected or positively selected.[2]

Where paired movers differ most from all workers is in their employers: they come from much larger and higher-paying firms. Selection on employer size gives us another way to characterize the nature of selection. We regress log earnings on log firm size (and year fixed effects) and compare each group's earnings with the level predicted by its firm size. Like all movers, paired movers are negatively selected. However, the extent of negative selection is smaller for paired movers than for all movers. Along many dimensions, then, paired movers are more similar to all workers than movers are. To the extent that researchers have been comfortable extrapolating from all movers to all workers, they should be at least as comfortable extrapolating from paired movers.

The paired movers come from larger firms and go to larger firms, which suggests that the dominant form of selection into paired mobility is simply chance. To assess the chance hypothesis, we take the origin firms and compute the empirical distribution of the destination firms of movers while not including the paired movers. Using these empirical probabilities, we then simulate mobility using the number of movers we see and compute the number of paired movers. Table 2 shows that this simple logic explains about 75% of the paired movers. This estimator understates the share because we omit the paired movers. Appendix Table A1 shows that at least a quarter of all paired movers are the only movers along the origin-destination-year path, and so cannot be generated in our simulation.

As another way of comparing paired movers to all movers, Table 3 shows the AKM decomposition in four samples. Columns (1) and (2) show the decomposition in the sample of all workers, where the columns differ by whether we use raw earnings or the residualized earnings, and show minimal differences. Column (3) reports the decomposition among all movers, and column (4) reports results for the paired movers. The adjusted-R^2—and thus the fit of the AKM model—is similar in the all-mover sample and the paired-mover sample. The samples differ in how much of the variance of earnings the firm effects explain: about 13% among all movers versus about 10% among paired movers.

Figure 1 shows that the distribution of earnings changes among the paired movers is broadly similar to that among all movers.

Veneto data: As in the Brazilian data the movers are negatively selected relative to all workers. The chance hypothesis only explains about 50% of the paired movers and they are more rare than in the Brazilian data.

3.1.2 Testing

Figure 2 shows the graphical version of the test for additive separability. The x-axis shows the gap in earnings between a pair of workers in period t, and the y-axis shows the gap in earnings in a subsequent year. The red circles show the first stage: the gap next year at the same firm. The

[2]They are slightly more negatively selected on education. Within-person education increases over time because of part-time school.

blue squares show the reduced form: the gap in the first year at the new firm. The first stage plot is approximately linear and very close to the 45-degree line, indicating that earnings gaps are quite stable. The reduced-form plot is also approximately linear, but the slope is much shallower than the first-stage slope. Hence, earnings gaps are not stable when workers move. The slope is below 1 because of the logic of Result 2: under uncorrelated match effects, the match effect gap is uncorrelated across firms so conditioning on the gap at one firm mechanically pulls the expected gap at the next firm towards zero.

Table 4 shows the main results for paired stayers and movers. Paired stayers provide a useful quantitative benchmark: their gaps cannot be unstable because of match effects, so any instability quantifies the role of drift in person effects or noise. Panel A shows the second-stage estimates. For the stayers, the paired-mover coefficient (the stayer coefficient) is 0.97, indicating minimal drift in person effects between periods. In contrast, for paired movers the coefficient is 0.76, which is meaningfully different from 1 and from the stayer coefficient.

Robustness to residualization: In our baseline results, we residualize for a polynomial in age interacted with gender and education. Appendix Table A2 shows that our results are similar if we instead just residualize for the age polynomial interacted with gender. As an alternative to residualizing earnings, the Table shows similar coefficients when matching on gender and 10-year age bins or gender and 1-year age bins. We focus on the residualized approach because it preserves sample sizes and parallels the treatment of covariates in the AKM literature.

Reweighting by firm size: One of the strongest forms of selection into the paired-mover sample is based on firm size. In Appendix Table A3 we reweight the paired movers so that either the origin firm size distribution matches the distribution among all workers, or the destination distribution does so. The point estimates are reasonably close: the baseline estimate is 0.76, reweighting based on origin firm size yields an estimate of 0.80, and reweighting the destination firm size yields 0.69.

Robustness to instrument: In Section 1, we discussed how we could relax Assumption 3(i) to allow an MA structure in the transitory error. Figure 3c shows that the paired-mover coefficient is fairly insensitive to which lag we use to instrument, rising from 0.76 to 0.80 as we use deeper lags.

Time series properties: In Section 1, we defined match effects to be a permanent feature of the match. Thus, up to transitory error the gap should be stable *within* the match. We already presented some evidence on this aspect of match effects by looking at paired stayers and finding a stayer coefficient of 0.97, which indicates small deviations from permanence. Moreover, the previous robustness exercise of varying the lag of the instrument is also indicative that the component we are capturing is quite permanent.

The correlation between the gaps at various horizons and in various subsamples of paired movers provides further evidence. We follow Lachowska et al. (2023, Footnote 9) and pool the autocorre-

lation coefficients using a regression.[3] For the full sample, we find an estimate of the persistence parameter of around 0.96 (see Appendix Table A4). This estimate compares to 0.976 that Lachowska et al. (2023, pg. 387) find for firm effects. Thus, the match effects are almost as persistent as firm effects.

Panel (a) of Figure 3 shows an event study representation of our results so far. To the left of the figure, we plot the stayer coefficient at each horizon within the origin firm; to the right, we instrument the gap at time 0 (the last period at the origin firm) with the gap at -1 and use the gap in each later year at the destination firm as the outcome. Two features stand out. First, the coefficient drops sharply at the move: gaps are much less stable across firms than within them. Second, on either side of the move the coefficient is roughly flat—it *declines* only gently at the destination—so the results do not depend on the year in which we measure the gap. Thus, the gap changes when workers move across firms but persists within firms, which is what we expect if match effects are a permanent feature of the match.

Reverse regression: For our baseline results, we ask whether earnings gaps at the origin firm predict gaps at the destination firm. Conceptually, we could do the reverse, and ask whether the gaps in the destination firm predict the gaps in the origin firm. Appendix C shows that our results are robust to reverse versions or using alternate periods.

Veneto data: In the Veneto data, the paired-mover coefficient is 0.82 versus 0.76. The stayer coefficient is 0.93 rather than 0.97.

3.1.3 Variance of match effects under exogenous mobility and uncorrelated match effects

Under exogenous mobility and uncorrelated match effects (Assumptions 1, 3, and 4), we find the variance of match effects is 0.027.[4] This estimate is about 40% of the variance of firm effects that we found in Table 3 for the full sample, which is a conservative comparison.[5]

Robustness to serial-correlation assumption: Assumption 3(i) separates match effects from the transitory error by requiring that the transitory error is serially uncorrelated within a match. As we discussed above, we can relax it to allow MA structures of various orders. Figure 3b shows that our estimates of the variance of match effects are somewhat sensitive to using lags up to the

[3] For completeness, we reproduce the logic here. Let $y_t = \phi_0 + \phi y_{t-1} + u_t$ so that the autocorrelation of y_t and y_{t-k} is $\rho(k) = \phi^k$. To estimate ϕ, fit $\ln(\rho(k)) = \theta k + \eta(k)$, where k is the lag order. The antilog of $\hat{\theta}$ is an estimate of ϕ. Table A5 shows the autocorrelation coefficients at various horizons, as well as the implied AR(1) coefficients. To compute the AR(1) coefficients we use the following logic. Let $\rho_{t,t+n}$ be the correlation of the gap between the paired workers at time t and $t+n$. Then under the $AR(1)$ model the persistence coefficient is given by $\frac{\rho_{t,t+2}}{\rho_{t,t+1}}$.

[4] $\widehat{\text{Var}}(\tilde{\mu}_{ij}|\mathcal{P}) = \frac{1}{2}(1-\hat{\beta}_1)\widehat{\text{Cov}}(Z_t^A, Z_{t-1}^A) = (1-0.758) \times 0.220 \times \frac{1}{2} = 0.027$, see equation (3) and Table 4.

[5] Our estimate of the variance of firm effects is 0.072 in the full sample, 0.065 in the full sample using residualized earnings, and 0.043 in the paired-mover sample. Gerard et al. (2021, Table 3) report estimates separately by demographic group and find variances of firm effects ranging from 0.056 to 0.086.

fifth (so making an MA(4) assumption on the transitory error). Specifically, the magnitude of the estimate of the variance of match effects declines from 0.027 to just under 0.020. This decline is driven more by the rise in the paired-mover coefficient than the decline in the gap covariance.[6] Even using the most conservative estimate, the magnitude is still over a quarter of the variance of firm effects. Thus, our estimate of the variance of match effects is reasonably robust given the range of serial-correlation assumptions that are feasible to implement. Moreover, given that median tenure at the origin firm is around two years, the estimator using one lag captures the persistence that is relevant for the experience of the median worker.

Dependence on size of the gap: One concern is that the paired-mover coefficient might be picking up on instability among pairs of workers with small gaps, while the larger gaps are more stable. In Appendix D we explore this possibility and find that in fact it is the larger gaps that are less stable. This finding implies that there are corners of the data with a much larger role for match effects than our baseline estimates.

3.1.4 Heterogeneity

The literature has sometimes been interested in the possibility that firm effects for men and women differ (e.g., Card, Cardoso, and Kline (2016)). Our approach would label these gender-specific firm effects as a form of match effects. If all of the match effects operated in this dimension, then this would be reassuring for our methods, but perhaps discouraging for the novelty of the finding. We are therefore interested in how much of our omnibus measure of match effects reflects gender-specific firm effects. We re-pair movers to ensure that both members of the pair are the same gender, and then re-estimate the variance of match effects among men and women separately. Table 4 shows that restricting the match effects to be within gender reduces the estimated magnitude of match effects by about 10% for women and 5% for men. Thus, over 90% of the omnibus match effects we detect are distinct from gender-specific firm pay premia.

Additional heterogeneity: In Appendix Table A6, we present numerous sample splits (Appendix Table A7 shows summary statistics for these sample splits). We split on the basis of whether workers changed sector and the pay at the firm. The general pattern is that moves in more similar contexts have more stable gaps, which is consistent with the idea that match effects are correlated within context such as sector. Appendix Table A8 reports the common finding that relative to a random benchmark workers are much more likely to stay in the same sector, which is consistent with violations of exogenous mobility. Appendix Table A9 splits the sample by whether the higher earner moved first. The gaps are more stable when the higher earner moved first.

[6] While equation (3) is not additive in its two estimated quantities, changing either $\hat{\beta}_1$ or $\widehat{\text{Cov}}(Z_t^A, Z_{t-k}^A)$ one at a time comes close to approximating the overall change. Using this approximation, over two-thirds of the decline is driven by the rise in the paired-mover coefficient.

3.2 Direct tests of exogenous mobility

So far we have presented results that depend on the assumption of exogenous mobility. As we developed in Section 1.4, under exogenous mobility the returns to predetermined characteristics do not systematically differ between the origin and destination firm. If we instrument for the earnings gap with predetermined characteristics, then deviations of the paired-mover coefficient from 1 offer a test of exogenous mobility.

In Table 5, we present this test for the gender gap, and also for differences in age and education. In all cases, the first stage is exceptionally strong, indicating that gaps in these characteristics predict earnings gaps at the origin firm. In all cases, the paired-mover coefficient is statistically significantly different from 1, indicating a rejection of exogenous mobility. The coefficients are especially far from 1 for age and gender. The evidence on gender offers an interesting parallel to the evidence in Table 4 that some of the match effects are about gender-specific firm effects. Paired movers of opposite genders tend to move to firms where the gaps in gender-specific firm effects are smaller. In contrast, if, as in Card, Cardoso, and Kline (2016), gender pay gaps were larger at higher-paying firms and opposite-gender workers systematically moved toward such firms, then we would expect a coefficient above 1.

As we showed in the last section, gender-specific returns are only a small portion of match effects. While the evidence presented in this section shows that there is endogenous mobility on this portion of match effects, it only captures part of potential endogenous mobility. Therefore, in the next section we study paired double movers.

Veneto data: The Veneto data does not have education and the age instrument is weak so we do not interpret the results. Similar to the Brazilian data, the gender instruments are strong and generate a rejection of exogenous mobility (though in the opposite direction).

3.3 Paired double movers: exogenous mobility and uncorrelated match effects

In the previous section, we presented direct evidence against exogenous mobility. We now build on this rejection by studying paired double movers. As we discussed in Section 1.5, paired *double* movers offer a test for the joint hypothesis of exogenous mobility and uncorrelated match effects.

3.3.1 Descriptive statistics on the sample of paired double movers

Table 1 shows statistics on the last year at firm A (column 5), last year at firm B (column 6) and first year at firm C (column 7). Paired double movers are positively selected on earnings relative to paired movers or all workers: their earnings are 4 log points higher than the paired movers at the first firm (interestingly, they are negatively selected on education). Consistent with the chance hypothesis, they also come from larger firms. Coming from larger firms partly explains why they make more: the prediction based on (log) size explains most of the difference at the first firm.

The last column of Table 3 shows the AKM decomposition in the sample of paired double movers. The variance of earnings is similar to the all-mover sample. The variance of person effects is slightly larger than among all movers, and the variance of firm effects is slightly smaller. The AKM model fits the data just as well in this sample as in the overall sample.

Figure 1b shows that relative to all movers the paired double movers have a distribution of earnings changes that is slightly compressed.

3.3.2 Testing

Table 6 shows the three IV estimates of the double-mover test for exogenous mobility and uncorrelated match effects. As discussed in Section 1, we follow the convention of reporting the IV estimate that is (weakly) below 1. The three IV estimates are 0.91, 0.92, and 0.99. Thus, there is clear evidence against the joint hypothesis of exogenous mobility and uncorrelated match effects.

In many models, match effects predict tenure. We split the paired double movers by the number of years they are observed at the middle firm (firm B). Figure 4 shows that two of the three IV estimates get smaller with more years at the middle firm.

4 A model of endogenous mobility

So far we have developed evidence against additive separability and exogenous mobility (by itself, and combined with uncorrelated match effects). We are now interested in estimating the variance of match effects while accounting for endogenous mobility, and in showing why match effects and endogenous mobility matter for applied questions. In Section 5, we use the model to revisit specification tests.

To do so, we write down and estimate a benchmark partial equilibrium search model. We chose this model because it maps closely to the paired-mover empirical moments. We suspect the applied lessons we develop in this model would carry over to richer models.

4.1 The model

The model is a standard partial equilibrium search model. Workers have a fixed effect $h_i \sim N(0, \sigma_h^2)$. Firms have a firm effect $p_j \sim N(0, \sigma_p^2)$. This distribution is also the offer distribution, denoted by F. The match effects distribution is $m_{ij} \sim N(0, \sigma_m^2)$. Denote this distribution by M. A worker's payoff to working at a firm j with firm effect p_j and match effect m_{ij} is given by $h_i + p_j + m_{ij}$.

The value of an employed worker with worker effect h_i at a firm with firm effect p_j and with match effect m_{ij} is:

$$W_i(p_j, m_{ij}) = h_i + p_j + m_{ij} + \frac{1}{1+r}\Bigg(\lambda_{1i}(1-\delta)\int_m\int_p \max\{W_i(p_j, m_{ij}), W_i(p', m')\}dF(p')dM(m') + (1-\lambda_{1i})(1-\delta)W_i(p_j, m_{ij}) + \delta U_i\Bigg).$$

An employed worker receives the flow payoff of her match. If her job is not destroyed, then she can receive an outside offer (which happens with probability λ_{1i}), and she makes a maximizing decision whether or not to accept it. The offer includes both the firm effect and the match effect and the worker considers both in her mobility decision. If she does not receive an offer (and her job is not destroyed), then she remains in her job. With probability δ her job is destroyed.

The fact that the arrival rate of offers on the job depends on the identity of the person—as in Lentz (2010)—gives the model a way of generating sorting between workers and firms. We parameterize the arrival rate as a linear function of the worker type: $\lambda_{1i} = \min\{\max\{0, a+bh_i\}, 1\}$, where the max and min operators are just ensuring that the arrival rate of offers on the job is bounded between 0 and 1.

The value function for an unemployed worker is standard and so is omitted.

The baseline model violates most of the assumptions we made in Section 1. It violates the second part of Assumption 1 (exogenous mobility) because workers select jobs on the basis of the match effect. It violates the second part of Assumption 2 (additive separability) because there are match effects. It also violates Assumption 4 (uncorrelated match effects) because workers select firms on the basis of match effects which induces correlation within worker across spells, across workers within spells, and across workers across spells in the match effects.

We also consider a version of the model where workers accept all offers. We term this model the exogenous mobility model. This version of the model only violates the second part of Assumption 2 (additive separability).

4.2 Estimation and fit

We estimate the model by the method of simulated moments. We assume that the flow value of unemployment is negative infinity so that all offers from unemployment are accepted. We simulate the model forward until steady state.

Estimation proceeds in two steps. In the first step, we estimate the EU rate (δ) and UE rate (λ_0) by matching quarterly flow rates computed in survey data. In the second step, we pick five parameters to match five moments. The five parameters are: the arrival rate of offers on the job (a and b), and the variance of person, firm and match effects (σ_h^2, σ_p^2, and σ_m^2). The five empirical moments are: the EE rate, the paired-mover coefficient, the variance of firm effects (estimated using Kline, Saggio, and Sølvsten (2020)), the variance of the earnings gap corrected for transitory error that we documented in Table 4, and the covariance of the firm and person effects. We search the parameter space using Metropolis-Hastings. Appendix E discusses how the model's parameters map into the moments.

Panel A of Table 7 shows that the model closely matches the five targeted moments. It misses one untargeted moment: the variance of person effects, which is about half as large in the model as in the data. The reason is that two targeted moments—the covariance between firm and person effects and the variance of the gap between paired movers—would mechanically determine the variance of person effects if paired movers were random draws of co-workers. They are not: paired

movers are more similar to each other than two random co-workers. The only channel the model has to generate this similarity is the dependence of the arrival rate of offers on person effects, but this mechanism is too weak. We target the gap between paired movers rather than the variance of person effects because the gap is an input to the exogenous mobility estimator of the variance of match effects. This choice makes it easier to compare the estimate and the model since the only difference is endogenous mobility.

We do not directly target the moments that feature in our test of exogenous mobility and uncorrelated match effects. Nevertheless, Table 7 shows that the model does a reasonable job of hitting the three IV coefficients. The direction of the miss is interesting: one reasonable prior is that a benchmark random search model would generate too much endogenous mobility relative to the data. Instead, it generates too little. In Appendix Table A10 we present an alternative calibration that targets these moments and below we show that our "applied" lessons are not dependent on which calibration we use.

4.3 Do match effects matter for applied questions?

The literature uses AKM models for two purposes. First, to estimate variance components. A prototypical example of this type of paper is Card, Heining, and Kline (2013). Second, to ask what would happen to earnings if we reallocated workers across firms. A prototypical example of this type of paper is Card, Cardoso, and Kline (2016), where the sorting component of the gender pay gap reflects this counterfactual thought experiment.

In the context of this model, the results in Table 7 suggest that the first type of exercise is slightly biased. Specifically, the estimate of the variance of firm effects is 0.065, and in the steady-state distribution the variance of the structural firm effects is 0.068. So the firm effects are biased down by less than 5%. There is much more bias in the match effects: the structural variance in the steady state is 0.034, while the exogenous mobility estimator finds a variance of 0.027. Relative to the truth, it is then biased down by 20%.

Panel (a) of Figure 5 echoes Borovičková and Shimer (2024) and shows that the second type of exercise is quite biased. We consider counterfactual moves to be the rejected offers in the model. We first ask how well the change in firm effects on these counterfactual moves predicts the hypothetical earnings changes if the moves had occurred. The intercept is -0.26 and the slope is 0.85. So the estimated firm effects do a poor job of predicting the change in earnings.

To make the bias concrete, consider moving workers from firms in the bottom 20% of estimated firm effects to firms in the top 20%. This exercise is a stylized version of the sorting component of the gender gap, which imagines a counterfactual in which women have the same distribution across firms as men and values the change at estimated firm effects. Here, we again use the rejected offers to ask about these counterfactual moves (see Appendix Table A11 for a tabular version of these results, as well as results using the double-mover calibration). The average change in the firm effects on rejected offers is 0.27, while the average change these workers would have received had they accepted their offers is -0.17. Thus, using firm effects for counterfactual questions in the

presence of match effects and endogenous mobility can be extremely biased.

5 Revisiting specification tests in the literature

So far we have developed evidence indicating a role for match effects. We have also developed evidence against exogenous mobility (both by itself, and in combination with uncorrelated match effects). This evidence is inconsistent with the results of three exercises in the literature that are supportive of both exogenous mobility and additive separability. In this section, we use a combination of the estimated model and the paired movers to revisit this evidence.

These specification tests point to real features of the data. This section asks two questions. Is the model of Section 4 consistent with these features of the data? And if so, why do tests meant to detect endogenous mobility or match effects fail to do so?

5.1 Event study test

The test that launched the modern Abowd, Kramarz, and Margolis (1999) literature is the seminal event study analysis in Card, Heining, and Kline (2013). The event study test asks whether the change in the firm effects (or mean co-worker earnings) predicts the change in earnings within worker.

Event study test in the model: In data simulated from the model in Section 4, we find an event study coefficient of 1.00.[7] This finding should be surprising given that the model we wrote down features endogenous mobility, and this test is supposed to detect endogenous mobility.

The test performs more intuitively when we split the sample into EE and EUE moves. EE moves give more scope to select on match effects than EUE moves and so should generate a lower event study coefficient (EUE moves are still selected on the origin firm match effect). Consistent with this intuition, Panel (b) of Figure 5 shows that on EE moves the coefficient is 0.54 and on EUE moves it is 1.04 (Appendix Table A13 shows regression versions). Thus, here the test works exactly as researchers expect and would detect that there are deviations from exogenous mobility on the EE moves. How is it that the test performs as we expect in subsamples, but not in the full sample?

To explain this result statistically, recall that a pooled regression combines regression coefficients in two subsamples as a function of three things: relative sample sizes, the within-group variances, and the between-group differences in sample means. The relative sample size and variance logic implies that the pooled coefficient should lie between the coefficients in the two samples. The difference in sample means can generate coefficients that lie outside the convex hull of the two subsamples and can generate unintuitive behavior; for example, Simpson's paradox occurs when the pooled coefficient is a different sign from the coefficient in each subsample. Here, the between-sample slope is above 1 and so pushes the pooled coefficient up to 1.

[7] See Appendix Table A12 for the event study test in the data.

Panel (c) of Figure 5 shows this between-sample logic most clearly. We raise the intercept in the arrival rate equation to 1.5, so that 97% of moves are EE moves. The coefficient is 0.49 in the EE sample, and 0.85 in the EUE sample, but the pooled coefficient is 0.92, above both. Appendix Figure A1 shows that the same results hold in the double-mover calibration.

We cannot prove that this pooling property holds for every form of endogenous mobility, but the logic is simple and should apply in most models of mobility. In our model, workers making EUE moves cannot select their destination on firm or match effects: the destination is a random draw from the offer distribution, while the origin was either a random draw or a firm reached by climbing the job ladder, so on average firm effects, match effects, and earnings all weakly fall. Workers making EE moves accept only offers that raise total earnings, and those gains come from some combination of higher firm effects and higher match effects. The EE sample therefore has more positive changes in both firm effects and earnings, so the between-sample slope is positive. It is above 1 because EE movers' earnings gains include the match-effect gain, while the x-axis measures only the change in firm effects.

This logic requires only that there be two kinds of mobility: one that maximizes income and one with less scope to do so. As evidence that it is not specific to our model, Appendix Table A14 applies the event study test to data simulated from the model in Borovičková and Shimer (2024), which is calibrated to match the Veneto results in Kline, Saggio, and Sølvsten (2020). The pooled coefficient is 0.94. Split by move type, it is 0.34 on EE moves and 0.93 on EUE moves. Both subsample coefficients lie below the pooled one, so the between-sample logic is what lets their model pass the event study test.

This between-sample logic explains how the same data can reject exogenous mobility using predetermined characteristics or paired double movers and yet pass the event study test. The event study test has low power against endogenous mobility in empirically plausible data, so its failure to reject is uninformative.

To our knowledge, this explanation for why the event study test can fail to reject is new. Bonhomme, Lamadon, and Manresa (2019) emphasize an explanation similar to the distinction between exogenous mobility and additive separability: even if there are match effects, if mobility is exogenous then the data will pass the event study test. Borovičková and Shimer (2024) emphasize that if the wage equation is additively separable, then the data will pass the event study test.[8]

5.2 Adjusted-R^2 in AKM and the interacted model

Card, Heining, and Kline (2013) show that the adjusted-R^2 only increases slightly between an additively separable model and one with match fixed effects, which is interpreted as evidence against the importance of match effects. Versions of this exercise are standard in the literature (e.g., Card, Heining, and Kline (2013, pg. 995) and Bonhomme, Lamadon, and Manresa (2019, pg. 718-9)). Table 3 shows that going from an additively separable model to a model with arbitrary

[8] By our test, however, data simulated from their model are not additively separable: the paired-mover coefficient is 0.64.

worker-firm interactions increases the adjusted-R^2 from 0.91 to 0.94, which implies a variance of match effects of 0.013 (in the paired-mover sample it is 0.019).[9] This variance of match effects is smaller than what the exogenous mobility estimator finds.

We use the model to assess this R^2-evidence. Holding fixed the parameters of the model, we simulate data under two assumptions on mobility and two panel lengths. The assumptions on mobility are exogenous mobility—where workers accept all offers—and endogenous mobility, which is the baseline model. The panel lengths are 100 periods and 7 periods, the number of periods in Card, Heining, and Kline (2013).

We report the adjusted-R^2 estimator of the variance of match effects,

$$\widehat{\text{Var}}(\text{match effects}) = (\hat{R}^2_{int} - \hat{R}^2_{AKM}) \times \text{Var}(y),$$

where $\hat{R}^2_{int}$ and $\hat{R}^2_{AKM}$ are the adjusted-R^2 from the model with worker-firm interactions and the AKM model, and $\text{Var}(y)$ is the overall variance of earnings. In the data we also multiply by $\hat{R}^2_{int}$ to adjust for noise, but in the model $\hat{R}^2_{int} = 1$, so the adjustment has no effect.

Table 8 shows that under exogenous mobility and long panels the standard adjusted-R^2 estimator is biased down. With endogenous mobility (but retaining the long panels), the adjusted-R^2 approach is further downward biased.

With short panels, the adjusted-R^2 logic is quite biased. With exogenous mobility it is biased down by over 50% and with endogenous mobility it is biased down by an order of magnitude: the implied variance of match effects is less than 10% of the truth.

Hence, in the presence of realistic data generating processes the adjusted-R^2 estimator is a downward biased estimator of the variance of match effects, which means that there is more scope for match effects than previously thought.

5.3 Bonhomme, Lamadon, and Manresa (2019)

Bonhomme, Lamadon, and Manresa (2019) (BLM) develop methods designed to detect deviations from additive separability. Do the paired movers just rediscover features of the data already documented by BLM? In this section, we show that the BLM approach does not reliably detect the deviations that paired movers detect.

The BLM model and estimation procedure: We consider the static model in BLM (see section 2.1 and equation (14) in BLM for details). In the static model, firms are assigned to classes, and workers are assigned to types. The static model in BLM parallels the AKM model in its key assumptions. Where the static model is richer than the AKM model is that it allows for interactions in the wage equation.

[9] $(0.937 - 0.910) \times 0.501 \times 0.937 = 0.013$, where the last 0.937 is a simple way of adjusting the variance of earnings for error. This calculation uses the full sample with residualized earnings. The calculation using only the paired movers is $(0.930 - 0.886) \times 0.454 \times 0.930 = 0.019$.

Estimation proceeds as in BLM. The key step is that we give the k-means clustering algorithm 30 percentiles of the log-earnings distribution among stayers. We use 10 firm clusters and 6 worker types. The static model implies worker-type specific transition probabilities across firm classes.

We compute the steady state distribution implied by these transition probabilities. From steady state, we simulate a set of moves. To construct wages, we draw from the error term. For the origin firm, we draw two error terms in order to be able to exactly replicate what we do in the data and instrument. In this simulated dataset, we then run the paired-mover regression and construct other statistics about the paired movers.

We estimate the BLM model in two samples. The first is all movers, which parallels the standard implementation of BLM. The second is only the paired movers, so that BLM and the paired-mover regression see exactly the same data.

For each sample we draw 100 random subsamples: 500,000 workers for all movers and 50,000 for paired movers. Estimating BLM on each subsample gives a distribution of BLM-implied paired-mover coefficients, which we compare with the coefficient in the data.

The paired-mover test in data based on the BLM model: Figure 6 shows that the data simulated from BLM estimates does not reflect the underlying patterns in the data and the direction of the bias cannot be signed. Panel (a) shows that the BLM-implied estimates of the paired-mover coefficient are biased down relative to the data. In contrast, Panel (b) shows that using only the paired movers the BLM-implied estimates of the paired-mover coefficient are biased up.

We interpret this evidence as indicative that the deviations from additive separability that the paired movers document are different from what the literature has previously found.

Monte Carlo evidence using the model: The evidence so far suggests that the BLM estimator is not able to reliably recover properties of the paired movers, which are the minimal feature of the data that speaks directly to the presence of match effects. To understand why this failure occurs, we simulate data from a version of the economic model in Section 4. We work with a discrete heterogeneity version of the model where the number of firm types and worker types matches that assumed in the BLM estimation procedure. For each worker-type-firm-type pair, we draw one permanent match effect. As above, we consider exogenous and endogenous mobility versions.

Appendix Figure A2 shows that BLM accurately recovers the paired-mover coefficient under exogenous but not endogenous mobility. The figure further shows that under either type of mobility the BLM estimator successfully recovers the wage equation. Where BLM struggles is that under endogenous mobility it misses on the mobility patterns: specifically, it finds that the destination firms are almost uniformly distributed among firm types, while in reality the higher firm types are more likely destinations.[10] Because the paired-mover coefficient and the gap variance are functions of the mobility patterns, missing on these leads to inaccurate estimates. Thus, BLM does not accurately recover the data underlying paired movers.

[10]This finding is consistent with the Appendix of BLM, which reports Monte Carlo evidence, but only about the wage equation and not mobility patterns.

6 Discussion

The idea of this paper is that paired movers, two workers who both move from firm A to firm B, are the simplest feature of the data that allows us to see match effects. Different instruments for the earnings gap at the origin firm test different hypotheses: whether match effects exist (additive separability) and whether worker mobility conditions on them (exogenous mobility). A third instrument tests a stronger hypothesis, exogenous mobility combined with uncorrelated match effects, which avoids taking a stance on the source of match effects.

We study these paired movers in administrative data and reject both additive separability and exogenous mobility. The earnings gap is much less stable across firms than within them, which rejects additive separability. Under exogenous mobility, this instability implies a variance of match effects that is about 40% of the variance of firm effects. We then reject exogenous mobility: the returns to gender, age and education differ systematically between the firms that paired movers leave and the firms they join. Paired double movers also reject the joint hypothesis of exogenous mobility and uncorrelated match effects.

Motivated by these results, we estimate a simple job search model that violates both additive separability and exogenous mobility. In the model, our exogenous mobility estimator understates the variance of match effects by about 20%, and the true variance is about half the variance of firm effects. We then ask whether two prototypical uses of AKM are biased. Variance decompositions are nearly unaffected: the (KSS-corrected) AKM estimate of the variance of firm effects is less than 5% below the structural variance. Counterfactual exercises are severely affected: for workers at bottom-quintile firms who reject an offer from a top-quintile firm, the change in firm effects predicts an earnings gain of 27 log points, while their earnings would in fact fall by 17.

These results run counter to the received wisdom about the AKM specification, so we revisit three pieces of evidence taken to support it. First, data generated with endogenous mobility can pass the event study test: the coefficient is well below 1 on job-to-job moves, but pooling them with moves through unemployment pushes it back to 1. Second, with endogenous mobility and short panels, the adjusted-R^2 comparison severely understates the variance of match effects. Third, the Bonhomme, Lamadon, and Manresa (2019) estimator does not reproduce the paired-mover coefficient, so it does not reliably detect the deviations from additive separability that paired movers reveal.

We see two avenues for future work. First, because match effects are large, understanding their structure and determinants, and how they shape mobility, is an important task. Second, movers designs are used well beyond wage equations (e.g., Finkelstein, Gentzkow, and Williams (2016), Chetty and Hendren (2018), and Lachowska, Sorkin, and Woodbury (2025)). The paired-mover sample could be informative in these contexts as well.

References

Abowd, John M, Francis Kramarz, and David N Margolis. 1999. "High wage workers and high wage firms." *Econometrica* 67 (2):251–333.

Beauregard, Pierre-Loup, Thomas Lemieux, Derek Messacar, and Raffaele Saggio. 2026. "Why do union jobs pay more? New evidence from matched employer-employee data." Tech. Rep. 33740, National Bureau of Economic Research.

Benedetto, Gary, John Haltiwanger, Julia Lane, and Kevin McKinney. 2007. "Using worker flows to measure firm dynamics." *Journal of Business & Economic Statistics* 25 (3):299–313.

Bonhomme, Stéphane, Thibaut Lamadon, and Elena Manresa. 2019. "A distributional framework for matched employer employee data." *Econometrica* 87 (3):699–739.

Borovičková, Katarína and Robert Shimer. 2024. "Assortative matching and wages: The role of selection." Tech. rep., National Bureau of Economic Research.

Card, David, Ana Rute Cardoso, Joerg Heining, and Patrick Kline. 2018. "Firms and labor market inequality: Evidence and some theory." *Journal of Labor Economics* 36 (S1):S13–S70.

Card, David, Ana Rute Cardoso, and Patrick Kline. 2016. "Bargaining, sorting, and the gender wage gap: Quantifying the impact of firms on the relative pay of women." *Quarterly Journal of Economics* 131 (2):633–686.

Card, David, Jörg Heining, and Patrick Kline. 2013. "Workplace heterogeneity and the rise of West German wage inequality." *Quarterly Journal of Economics* 128 (3):967–1015.

Chetty, Raj and Nathaniel Hendren. 2018. "The impacts of neighborhoods on intergenerational mobility I: Childhood exposure effects." *Quarterly Journal of Economics* 133 (3):1107–1162.

Eeckhout, Jan and Philipp Kircher. 2011. "Identifying sorting—in theory." *The Review of Economic Studies* 78 (3):872–906.

Finkelstein, Amy, Matthew Gentzkow, and Heidi Williams. 2016. "Sources of geographic variation in health care: Evidence from patient migration." *Quarterly Journal of Economics* 131 (4):1681–1726.

Gerard, François, Lorenzo Lagos, Edson Severnini, and David Card. 2021. "Assortative matching or exclusionary hiring? The impact of employment and pay policies on racial wage differences in Brazil." *American Economic Review* 111 (10):3418–3457.

Gibbons, Robert, Lawrence F Katz, Thomas Lemieux, and Daniel Parent. 2005. "Comparative advantage, learning, and sectoral wage determination." *Journal of Labor Economics* 23 (4):681–724.

Hagedorn, Marcus, Tzuo-Hann Law, and Iourii Manovskii. 2017. "Identifying equilibrium models of labor market sorting." *Econometrica* 85 (1):29–65.

Kantenga, Kory and Tzuo-Hann Law. 2016. "Sorting and wage inequality." Tech. rep., University of Pennsylvania. URL `https://red-files-public.s3.amazonaws.com/meetpapers/2016/paper_660.pdf`.

Kline, Patrick. 2024. "Firm wage effects." *Handbook of Labor Economics* 5:115–181.

Kline, Patrick, Raffaele Saggio, and Mikkel Sølvsten. 2020. "Leave-out estimation of variance components." *Econometrica* 88 (5):1859–1898.

Lachowska, Marta, Alexandre Mas, Raffaele Saggio, and Stephen A Woodbury. 2023. "Do firm effects drift? Evidence from Washington administrative data." *Journal of Econometrics* 233 (2):375–395.

Lachowska, Marta, Alexandre Mas, and Stephen A Woodbury. 2020. "Sources of displaced workers' long-term earnings losses." *American Economic Review* 110 (10):3231–3266.

Lachowska, Marta, Isaac Sorkin, and Stephen A Woodbury. 2025. "Employers and unemployment insurance take-up." *American Economic Review* 115 (8):2529–2573.

Lentz, Rasmus. 2010. "Sorting by search intensity." *Journal of Economic Theory* 145 (4):1436–1452.

Sorkin, Isaac. 2018. "Ranking firms using revealed preference." *Quarterly Journal of Economics* 133 (3):1331–1393.

Woodcock, Simon D. 2015. "Match effects." *Research in Economics* 69 (1):100–121.

Table 1: Summary statistics

	All workers	All movers	Paired movers		Paired double movers		
			Firm A	Firm B	Firm A	Firm B	Firm C
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
A. Brazil							
Avg. age	34.09	30.90	33.24	34.32	33.29	35.01	36.03
Share female	0.40	0.35	0.31	0.31	0.27	0.27	0.27
Years of schooling	11.30	11.10	10.81	11.06	10.19	10.47	10.65
Avg. Tenure (months)	46.97	23.27	39.40	11.29	25.86	20.80	12.37
Median Tenure (months)	24.74	13.52	28.36	10.34	15.75	14.41	10.13
Avg. log earnings	2.34	2.19	2.39	2.40	2.43	2.51	2.53
Predicted log earnings (firm size)	2.34	2.32	2.47	2.48	2.52	2.56	2.56
# of Worker-Years	555,346,010	79,856,352	6,772,958	6,772,958	1,243,568	1,243,568	1,243,568
# of Workers	84,457,512	39,416,039	5,913,103	5,913,103	1,099,382	1,099,382	1,099,382
Avg. firm size	12.10	25.27	195.82	195.92	461.85	453.47	467.54
Avg. firm pay	2.34	2.35	2.45	2.48	2.41	2.47	2.52
# of Firm-Years	46,405,377	18,220,954	964,673	1,033,253	164,105	162,519	170,802
# of Firms	8,098,007	5,362,858	453,191	493,446	96,615	94,340	98,827
Var(Z) (earnings gap)	0.404	0.370	0.308	0.253	0.234	0.217	0.259
B. Veneto							
Avg. age	34.17	30.14	35.44	36.44			
Share female	0.37	0.37	0.37	0.37			
Avg. log earnings	10.10	9.64	10.18	10.16			
# of Worker-years	15,158,383	1,897,250	20,892	20,892			
# of Workers	2,156,378	984,857	20,223	20,223			
Avg. firm size	8.35	15.79	222.27	237.15			
Avg. firm pay	10.10	10.11	10.17	10.24			
# of Firm-years	1,815,553	676,890	6,607	6,883			
# of Firms	264,742	202,672	3,904	3,876			
Var(Z) (earnings gap)	1.07	1.01	0.399	0.600			

This table shows descriptive statistics of the Brazilian (Panel A) and Veneto (Panel B) data, described in Section 2. Column 1 displays all workers in the data and reports statistics over all employment spells. Column 2 shows all workers who move firms and reports statistics of the spell prior to the move (origin firm). Columns 3 and 4 show workers in the paired-mover sample, both in their last year in the origin firm (Column 3) and in their first year in the destination firm (Column 4). Finally, Columns 5–7 show workers in the sample of paired double movers, both in their last year in the first and middle firms (Columns 5–6), and in their first year in the third firm (Column 7). Panel A uses the universe of spells in RAIS from 2007 to 2022, while Panel B uses Veneto Worker History data from 1985 to 2001. We do not report statistics for paired double movers in Veneto because the sample is too small. Firm-level statistics are computed using all workers in the firm, not only (paired/double) movers. "Predicted log earnings (firm size)" is the predicted log earnings from a regression of log earnings on log firm size and year fixed effects. Var(Z) is the earnings gap between pairs of workers. For all workers and all movers we form as many pairs as possible using just information in the origin firms.

Table 2: Paired movers *vs.* what would be predicted by chance

	Paired movers
A. Brazil	
Data	3,386,479
Simulation (Mean)	2,541,448
Simulation (p5)	2,540,787
Simulation (p95)	2,542,076
B. Veneto	
Data	10,446
Simulation (Mean)	5,240
Simulation (p5)	5,209
Simulation (p95)	5,268

This table shows the number of paired movers that would occur by chance given the mobility patterns in the data. We take the sample of movers that could be paired and drop those in the corresponding final sample. For each origin firm, we compute the distribution of destination firms over a one-year period, pooling all years in the data. We use these distributions to simulate a sample of movers of the same size as in the data, and then create a paired-mover sample. Panels A (Brazil) and B (Veneto) show the number of simulated paired movers across 100 draws.

Table 3: AKM estimates (KSS-corrected)

	All workers (1)	All workers* (2)	All movers* (3)	Paired movers* (4)	Paired double movers* (5)
A. Brazil, 2012–2017					
Var(Y)	0.506	0.501	0.442	0.454	0.442
Var(α)	0.283	0.294	0.245	0.292	0.284
Var(ψ)	0.072	0.065	0.059	0.043	0.041
Cov(ψ, α)	0.050	0.047	0.040	0.033	0.031
Var(ε)	0.051	0.048	0.058	0.053	0.054
R2	0.929	0.931	0.901	0.911	0.909
Adj. R2	0.907	0.910	0.869	0.886	0.879
Adj. R2, int.	0.934	0.937	0.924	0.930	0.928
Observations	40,950,644	40,950,644	22,599,194	2,105,768	310,015
# Firms	911,755	911,755	911,755	62,776	11,705
# Workers	8,764,893	8,764,893	4,598,015	411,945	65,674
# Movers	4,598,015	4,598,015	4,598,015	411,945	65,674
# Moves	14,585,164	14,585,164	14,585,164	1,251,210	265,280
B. Veneto, 1996–2001					
Var(Y)	0.766	0.714	0.697	0.808	
Var(α)	0.317	0.305	0.241	0.416	
Var(ψ)	0.129	0.104	0.091	0.059	
Cov(ψ, α)	0.041	0.039	0.038	0.063	
Var(ε)	0.238	0.227	0.289	0.207	
R2	0.820	0.816	0.714	0.815	
Adj. R2	0.753	0.746	0.618	0.754	
Adj. R2, int.	0.797	0.797	0.759	0.810	
Observations	5,111,991	5,111,991	1,702,650	15,975	
# Firms	61,638	61,638	61,638	615	
# Workers	1,332,729	1,332,729	363,956	3,325	
# Movers	363,956	363,956	363,956	3,325	
# Moves	988,088	988,088	988,088	8,334	

This table shows the results of the AKM estimation with the KSS correction in Brazilian (Panel A) and Veneto (Panel B) data (Abowd, Kramarz, and Margolis, 1999; Kline, Saggio, and Sølvsten, 2020). The asterisk ($*$) indicates residualized earnings. Columns 1 and 2 show the results for all workers. In Brazil, we use a 25% random sample of workers followed from 2012–2017, while in Veneto we use all work spells from 1996–2001. Column 3 restricts the samples only to workers that move at least once in the period, and Columns 4 and 5 further restrict the sample to workers in the paired and paired double movers samples, respectively (and whose associated move(s) in the paired samples are in the same period as the AKM estimation). We report the KSS-corrected variance terms, the R^2 and adjusted R^2 of the AKM model, and the adjusted R^2 of a regression of the log earnings on fully interacted person and firm fixed effects. The last five rows show the number of observations, firms, workers, movers, and moves in the leave-one-out connected set used in estimation.

Table 4: The paired-mover coefficient

			Same-gender pairs	
	Baseline	Stayers	Men	Women
	(1)	(2)	(3)	(4)
A. Brazil, 2nd Stage				
Gap in Origin	0.758	0.969	0.746	0.769
	(0.0005)	(0.0002)	(0.0007)	(0.0009)
$\text{Cov}(Z_t^A,\, Z_{t-1}^A)$	0.220	0.485	0.202	0.207
$\text{Var}(\tilde{\mu}_{ij} \mid \mathcal{P})$	0.027	0.008	0.026	0.024
Observations	3,386,479	3,144,821	2,129,394	894,247
B. Brazil, 1st Stage				
Gap in Origin (lag)	0.918	0.949	0.913	0.918
	(0.0004)	(0.0002)	(0.0005)	(0.0007)
F-statistic	6,473,532.2	34,159,725.6	3,534,077.3	1,970,069.2
C. Veneto, 2nd Stage				
Gap in origin	0.816	0.929	0.792	0.818
	(0.024)	(0.005)	(0.037)	(0.054)
$\text{Cov}(Z_t^A,\, Z_{t-1}^A)$	0.209	0.234	0.154	0.177
$\text{Var}(\tilde{\mu}_{ij} \mid \mathcal{P})$	0.019	0.008	0.016	0.016
Observations	10,446	96,981	5,825	3,158
D. Veneto, 1st Stage				
Gap in Origin (lag)	0.427	0.498	0.388	0.353
	(0.008)	(0.002)	(0.011)	(0.014)
F-statistic	3,021.6	56,582.9	1,254.2	657.4

This table shows the paired-mover coefficient (i.e., the coefficient on the gap in the origin). We estimate a TSLS regression where the earnings gap in the destination firm is regressed on the earnings gap in the origin firm, instrumented by the lagged earnings gap in the origin firm. Panels A and B (C and D) show the second and first stages of the test in Brazilian (Veneto) data, respectively. Column 1 shows the baseline estimate using the full paired-mover sample (Columns 3–4, Table 1). Column 2 shows the estimate using a benchmark sample of paired stayers. Columns 3 and 4 show the estimate using a paired-mover sample in which both workers are of the same gender, running separate regressions for men and women. Finally, we also report the covariance of the gap in the origin, and the variance of match effects under exogenous mobility calculated using Equation 3.

Table 5: Direct test for exogenous mobility: alternative instruments

	Gender (1)	Age (2)	Education (3)
A. Brazil, 2nd Stage			
Gap in Origin	0.884	1.29	0.957
	(0.004)	(0.005)	(0.002)
Observations	3,386,479	3,386,479	3,386,479
B. Brazil, 1st Stage			
W1 female	-0.131		
	(0.001)		
W2 female	0.128		
	(0.001)		
W1 female × W2 female	0.003		
	(0.002)		
Gap in age		-0.006	
		(0.0001)	
Gap in years of education			0.048
			(0.0001)
F-statistic	13,309.9	50,919.2	233,601.4
C. Veneto, 2nd Stage			
Gap in Origin	1.14	-7.36	
	(0.050)	(13.0)	
Observations	10,446	10,446	
D. Veneto, 1st Stage			
W1 female	-0.382		
	(0.021)		
W2 female	0.354		
	(0.021)		
W1 female × W2 female	0.048		
	(0.030)		
Gap in age		0.001	
		(0.001)	
F-statistic	246.1	0.369	

This table shows the paired-mover coefficient using alternative instruments for the gap in earnings at the origin firm. The sample is the same as in Column 1 of Table 4. Column 1 uses the full interaction of both workers' gender as instruments, Column 2 uses the gap in age, and Column 3 uses the gap in years of education. Panels A and B (C and D) show the second and first stages of the test in Brazilian (Veneto) data, respectively. There is no education variable in the Veneto data.

Table 6: Paired double movers

Instrument:	Z^A	Z^B	Z^C
Outcome:	Z^C	Z^C	Z^A
Endogenous variable:	Z^B	Z^A	Z^B
	(1)	(2)	(3)
A. 2nd Stage			
Z^B	0.908		0.922
	(0.002)		(0.002)
Z^A		0.985	
		(0.002)	
Observations	621,784	621,784	621,784
B. 1st Stage			
Z^A	0.582		
	(0.001)		
Z^B		0.628	
		(0.001)	
Z^C			0.517
			(0.001)
F-statistic	357,620.3	357,620.3	291,979.3
Cov(Z^A, Z^B)	0.136	0.136	
Cov(Z^B, Z^C)		0.134	0.134
Cov(Z^A, Z^C)	0.124		0.124

This table shows three versions of the double-mover regression. We estimate a TSLS regression where the earnings gap in firms A (Z^A), B (Z^B), and C (Z^C) may play the role of the dependent variable, endogenous regressor, or instrument. As described in Section 1, we report the three specification where $\hat{\beta}_1^{\text{double}} < 1$. Panels A and B show the second and first stages, respectively. In each column, we also report the corresponding covariances, whose ratios give the IV coefficients. The sample of paired double movers is shown in Columns 5-7 of Table 1.

Table 7: Model results

	(1)	(2)	(3)	(4)	(5)	(6)
A. Model fit						
	EE rate	β_1	Cov(Z_t,Z_{t-1})	Var(ψ_j)	Var(α_i)	Cov(α_i, ψ_j)
Empirical Moments (Data)	0.038	0.758	0.220	0.065	0.294	0.047
Simulated Moments (Model)	0.037	0.757	0.215	0.065	0.144	0.046
B. Model parameters						
	λ_0 (UE rate)	δ (EU rate)				
Calibrated Parameters	0.072	0.076				
	σ_m^2	σ_h^2	σ_p^2	a	b	
Estimated Parameters	0.030	0.089	0.064	0.140	1.171	
C. Structural moments (steady-state)						
	m_{ij}	h_i	p_j			
Mean	0.083	0.047	0.132			
Variance	0.034	0.096	0.068			
Cov(h_i, p_j)	0.038					
Cov(h_i, m_{ij})	0.022					
Cov(p_j, m_{ij})	0.008					
D. Double-mover moments						
	Cov(Z^A, Z^B)	Cov(Z^B, Z^C)	Cov(Z^A, Z^C)	β_1	β_2	β_3
Data	0.136	0.134	0.124	0.908	0.922	0.985
Model	0.128	0.132	0.133	0.968	0.969	0.998
E. Implied variance of match effects under exogenous mobility						
	Var($\tilde{\mu}_{ij} \mid \mathcal{P}$)					
Data	0.027					
Model	0.026					

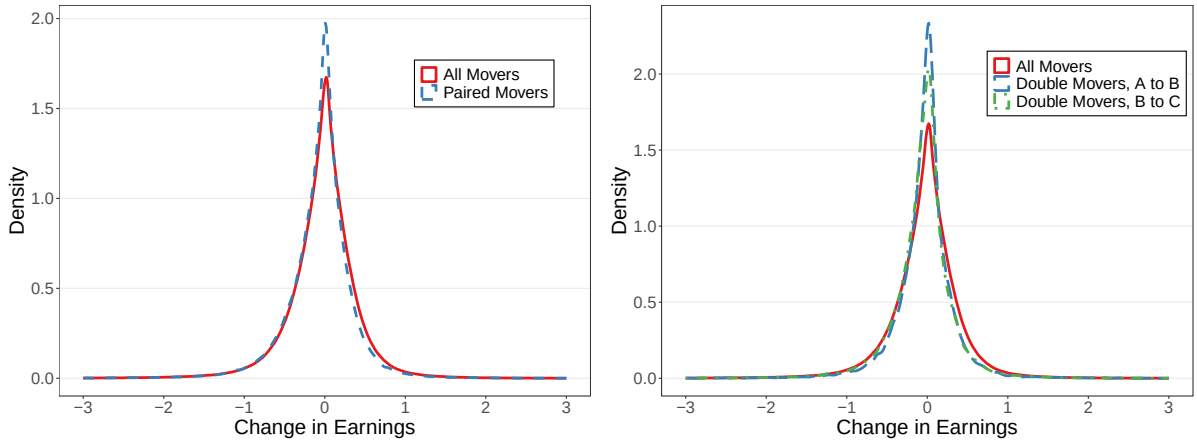
This table shows the model calibration. We fix the unemployed arrival rate λ_0 and the separation rate δ to quarterly UE and EU rates in Brazil, measured using labor force surveys (PNAD). We estimate the five remaining parameters of the model, the on-the-job arrival rate $\lambda_{1i} = \min\{\max\{0, a + b \times h_i\}, 1\}$, the variance of match effects σ_m^2, the variance of person effects σ_h^2, and the variance of firm effects σ_p^2, to match: the EE rate, the paired-mover coefficient β, the gap covariance at the origin firm Cov(Z_t, Z_{t-1}), the variance of firm effects $Var(\psi_j)$, and the covariance of firm and person effects Cov(α_i, ψ_j); Var(α_i) and the double-mover moments are not targeted. We simulate 200,000 workers and 200 firms and pull 100 periods out of the steady-state distribution of the model to compute the moments.

Table 8: Match effects estimators under different mobility assumptions

	Long Panel ($T = 100$)		Short Panel ($T = 7$)	
	Exogenous (1)	Endogenous (2)	Exogenous (3)	Endogenous (4)
A. Structural variances				
Var(Y)	0.190	0.348	0.190	0.350
Var(h_i)	0.097	0.097	0.097	0.097
Var(p_j)	0.064	0.078	0.064	0.079
Var(m_{ij})	0.030	0.033	0.030	0.033
B. Variance components estimates				
Var(α_i) - AKM	0.102	0.157	0.116	0.183
Var(ψ_j) - AKM	0.064	0.072	0.064	0.062
Var(α_i) - KSS	0.097	0.142	0.110	0.180
Var(ψ_j) - KSS	0.064	0.072	0.064	0.062
C. Intermediate statistics				
R2	0.873	0.947	0.947	0.994
Adj. R2	0.871	0.946	0.928	0.992
Adj. R2, interactions	1.000	1.000	1.000	1.000
D. Implied variance of match effects				
Var(μ_{ij}) - Adj. R2	0.025	0.019	0.014	0.003

This table shows the performance of match effects estimators with different panel lengths and mobility assumptions. We simulate data from the model in Table 7 under two scenarios: exogenous mobility, where workers accept any outside offer, and endogenous mobility (baseline). We draw two panel lengths from the model steady state: a long panel with 100 periods (Columns 1 and 2) and a short panel with 7 periods (Columns 3 and 4). Panel A shows structural parameters: the true variances of wages, person effects, firm effects, and match effects. Panel B shows the estimates of the variance of person and firm effects. Panel C shows intermediate statistics: the (adjusted) R^2 of the two-way fixed effects regression, and the adjusted R^2 of a fixed effects regression with worker-firm interactions. Finally, Panel D shows the variance of match effects implied by the adjusted R^2, $(1 - \text{Adj. } R^2) \times \text{Var}(Y)$. The results in each column are averaged over 200 simulations.

Figure 1: Change in earnings around the move, Brazil

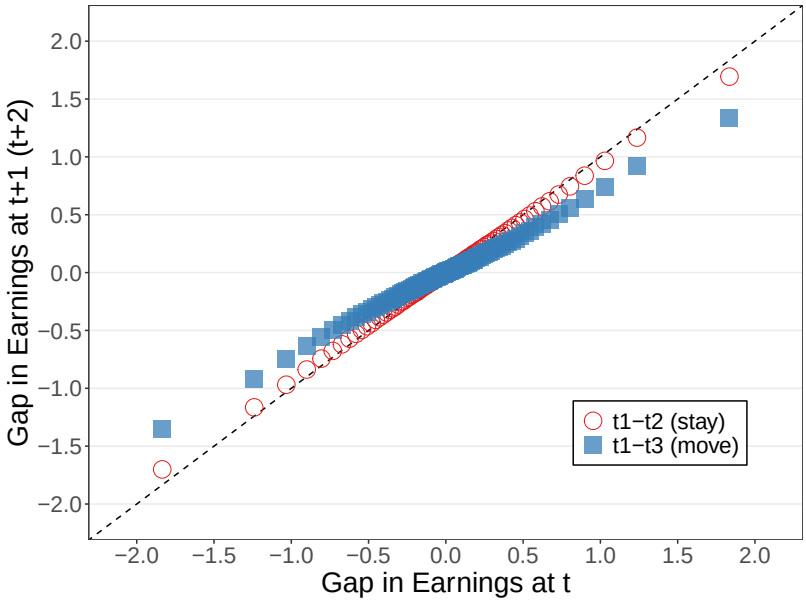


(a) Paired movers

(b) Paired double movers

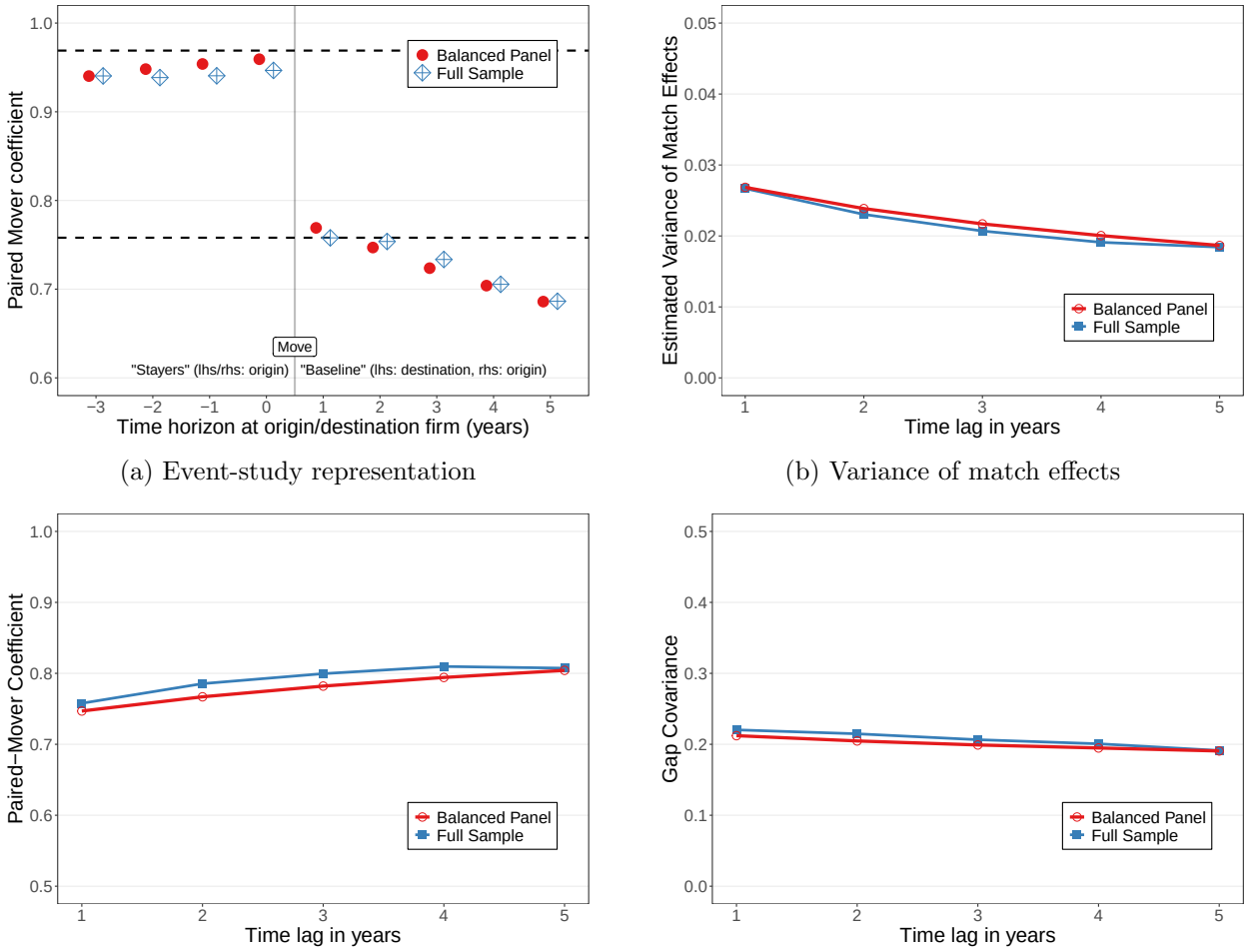
This figure shows the distribution of the change in log earnings between the destination and origin firms for all movers (Column 2, Table 1) and the final sample of paired movers (Columns 3-4, Table 1) in Panel A and paired double movers (Columns 5–7, Table 1) in Panel B.

Figure 2: Paired movers: changes in the earnings gap



This figure shows the graphical version of the paired-mover coefficient. The x-axis shows the gap in earnings in the second-to-last period in the origin firm. We split the gap in earnings into 100 bins. The red circles show the first stage, so the y-axis represents earnings gaps in the last period at the origin firm. The blue squares show the reduced-form, so the y-axis represents earnings gaps in the first period at the destination firm. The dashed line shows the 45-degree line.

Figure 3: Temporal stability



(a) Event-study representation

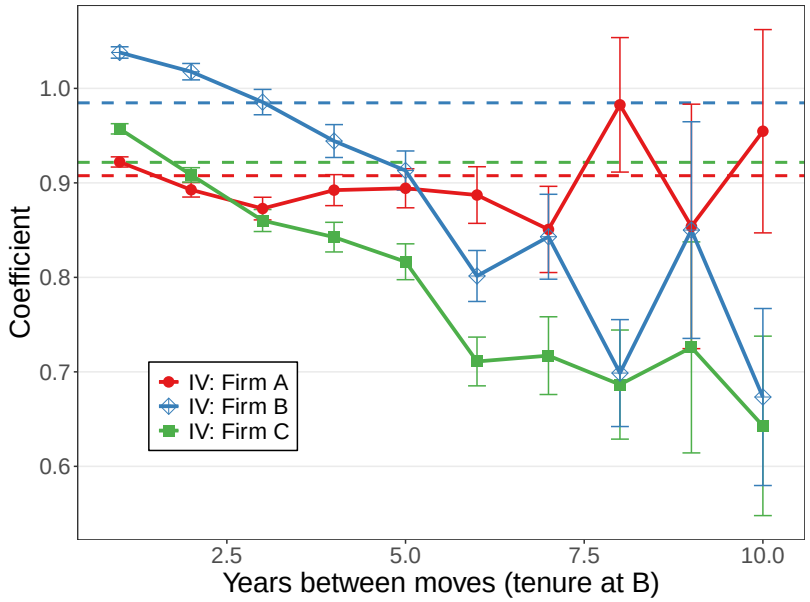
(b) Variance of match effects

(c) Paired-mover coefficient

(d) Gap covariance

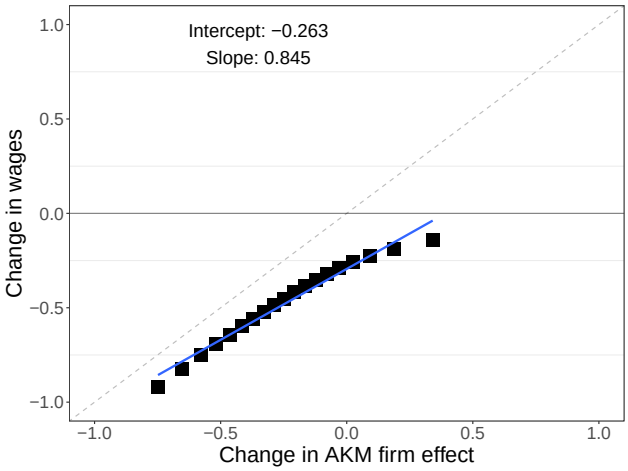
This figure shows temporal stability of the paired-mover coefficient. In Panel A, the lower dashed horizontal line shows the baseline movers estimate from Column 1 of Table 4. The higher dashed horizontal line shows the baseline stayers estimate from Column 2 of Table 4. The gray vertical line marks the timing of the move between origin and destination firms. To the left of the vertical line, we report the paired-mover coefficient for stayers at different horizons. To the right of the vertical line, we report the paired-mover coefficient where the first stage is always the last two years at the origin firm and the outcome rolls forward in time. In Panel B, we show the variance of match effects under exogenous mobility, calculated using the estimator from Equation 3. The x-axis shows different horizons (lags) used to compute the estimator's components: the paired-mover coefficient, where the gap at the origin firm Z_t is instrumented with $Z_{t-1}, Z_{t-2}, \ldots$, and the covariance of the gap at the origin firm, $Cov(Z_t, Z_{t-1}), Cov(Z_t, Z_{t-2}), \ldots$. Panels C and D show the paired-mover coefficient and the covariance of the gap at different lags used in Panel B. In all panels, we show estimates using the full sample (all paired movers observed at that period) and a balanced panel (only paired movers observed at all periods).

Figure 4: Paired double movers

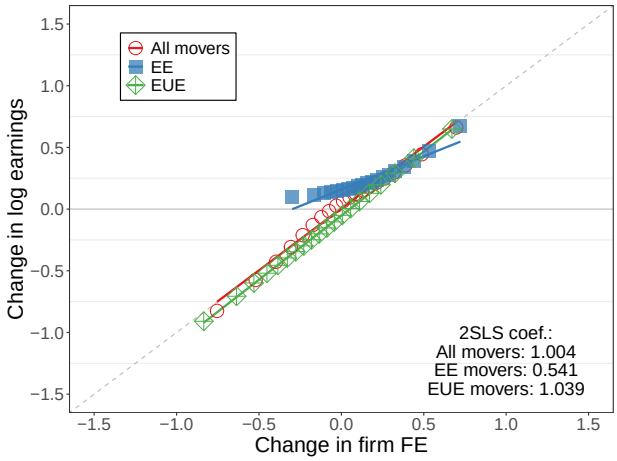


This figure shows the double-mover regression. We report the 2nd stage coefficient according to different tenure lengths at the middle firm (firm B). As described in Section 1, we report the three possible specifications with $\hat{\beta} < 1$ in the full sample. The pooled coefficients using the full sample are shown by the horizontal dashed lines. Across all regressions, tenure at firms A and C is unrestricted.

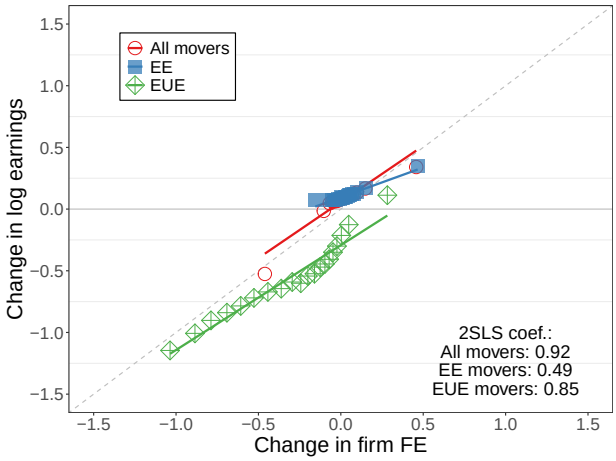
Figure 5: Model simulated data



(a) AKM firm effects predictions for rejected offers



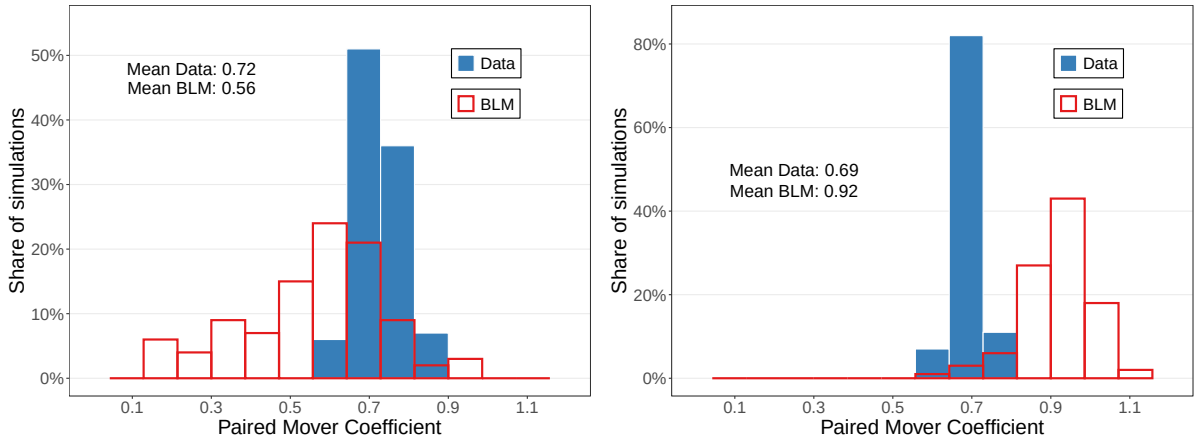
(b) Event study, 43% EE movers ($a = 0.14$)



(c) Event study, 97% EE movers ($a = 1.50$)

This figure shows results using data simulated from the model. In Panel A, we compare the change in firm effects (x-axis) to the change in wages (y-axis) for rejected offers. Panels B and C show the event study test in different samples. We plot the change in wages between destination and origin firms against the change in firm effects. Panel B uses the baseline on-the-job arrival rate parameter $a = 0.14$, while Panel C sets $a = 1.50$.

Figure 6: Paired movers and BLM



(a) All movers

(b) Paired movers

This figure shows the distribution of the paired-mover coefficient in the data and in data simulated from the estimated BLM model across 100 iterations. In Panel A, in each run we sample 500,000 workers from all movers (Column 3 of Table 1). We then compute the paired-mover coefficient using this sample, estimate BLM on the same sample, and simulate data from the estimated BLM model to obtain the paired-mover coefficient in the simulated data. In Panel B, we repeat the same exercise but we sample 50,000 workers from the sample of paired movers (Column 4 of Table 1).

Supplementary appendix to Paired movers

Isaac Sorkin and Lucas Warwar

A Omitted proofs

The results in this paper are statements about the covariance between the earnings gaps of a pair of workers in two different periods. This Appendix first proves two lemmas about any such covariance. Then we prove the results stated in the text as special cases.

A.1 Preliminaries

Notation. Let $\mathcal{Q}$ denote the relevant paired-mover population: $\mathcal{P}$ for paired movers and $\mathcal{P}_2$ for paired double movers. For a pair $(i, i') \in \mathcal{Q}$ and a firm j at which both workers are observed, write

$$\Delta\alpha \equiv \alpha_i - \alpha_{i'}, \qquad \Delta\mu^j \equiv \mu_{ij} - \mu_{i'j}, \qquad \nu_t^j \equiv \tilde{\epsilon}_{ijt} - \tilde{\epsilon}_{i'jt}, \tag{A1}$$

so that $Z_j^t = \Delta\alpha + \Delta\mu^j + \nu_t^j$, where we suppress the pair subscripts for simplicity. Define the second moments

$$\kappa_j \equiv \mathrm{Cov}(\Delta\alpha, \Delta\mu^j|\mathcal{Q}), \qquad \omega_{jk} \equiv \mathrm{Cov}(\Delta\mu^j, \Delta\mu^k|\mathcal{Q}), \tag{A2}$$

so that $\omega_{jj} = \mathrm{Var}(\Delta\mu^j|\mathcal{Q})$. κ_j is the covariance between the person-effect gap and the match-effect gap at firm j while ω_{jk} with $j \neq k$ is the covariance of the match-effect gaps across two firms. A useful combination of these variances and covariances is

$$K_{jk} \equiv \mathrm{Var}(\Delta\alpha|\mathcal{Q}) + \kappa_j + \kappa_k + \omega_{jk}. \tag{A3}$$

We use the following two facts repeatedly.

Fact 1 (Measurability)**.** *Let $\mathcal{C}$ be any conditioning set that contains $\mathcal{G}$, such as $(\mathcal{G}, \alpha, \psi, X)$ or $(\mathcal{G}, \alpha, \psi, X, \mu)$. The events $\{(i, i') \in \mathcal{P}\}, \{(i, i') \in \mathcal{P}_2\}$ and $\{j(i, t) = j\}$ are functions of $\mathcal{G}$. Therefore, if $\mathbb{E}[V|\mathcal{C}] = 0$, the law of iterated expectations gives $\mathbb{E}[V|\mathcal{Q}] = 0$ and $\mathbb{E}[WV|\mathcal{Q}] = 0$ for every W that is a function of $\mathcal{C}$. Specifically,* $\mathrm{Cov}(W, V|\mathcal{Q}) = 0$.

Fact 2 (Random ordering)**.** *Within each pair the labels i and i' are assigned at random, so the labeled pairs (i, i') and (i', i) are equally likely. Hence, conditional on $\mathcal{Q}$, $\Delta\alpha$, $\Delta\mu^j$, ν_t^j, and Z_t^j are symmetrically distributed about zero and have mean zero, and* $\mathrm{Var}(\mu_{ij}|\mathcal{Q}) = \mathrm{Var}(\mu_{i'j}|\mathcal{Q})$.

A.2 Covariance lemmas

Lemma A1 (Cross-spell covariances)**.** *Let $(i, i') \in \mathcal{Q}$ be observed at firm j in period t and at firm k in period s, with $t \neq s$. Suppose the first part of Assumption 1 and Assumption 3(ii) holds; if*

$j = k$ *suppose also that Assumption 3(i) holds. Then*

$$\mathrm{Cov}(Z_t^j, Z_s^k|\mathcal{Q}) = \mathrm{Var}(\Delta\alpha|\mathcal{Q}) + \kappa_j + \kappa_k + \omega_{jk} = K_{jk}. \tag{A4}$$

Proof. We expand the covariance:

$$\begin{aligned}\mathrm{Cov}(Z_t^j, Z_s^k|\mathcal{Q}) = \mathrm{Var}(\Delta\alpha|\mathcal{Q}) + \mathrm{Cov}(\Delta\alpha, \Delta\mu^j|\mathcal{Q}) + \mathrm{Cov}(\Delta\alpha, \Delta\mu^k|\mathcal{Q}) + \mathrm{Cov}(\Delta\mu^j, \Delta\mu^k|\mathcal{Q}) \\ + \mathrm{Cov}(\Delta\alpha + \Delta\mu^j, \nu_s^k|\mathcal{Q}) + \mathrm{Cov}(\Delta\alpha + \Delta\mu^k, \nu_t^j|\mathcal{Q}) + \mathrm{Cov}(\nu_t^j, \nu_s^k|\mathcal{Q}).\end{aligned} \tag{A5}$$

The first line equals K_{jk} by definition. It remains to show that the three terms on the second line are zero.

Signal-noise terms (first two terms). By the first part of Assumption 1, $\mathbb{E}[\nu_s^k|\mathcal{G}, \alpha, \psi, X, \mu] = 0$. Note that $\Delta\alpha + \Delta\mu^j$ is a function of (α, μ). By Fact 1, $\mathbb{E}[\nu_s^k|\mathcal{Q}] = 0$ and $\mathbb{E}[(\Delta\alpha + \Delta\mu^j)\nu_s^k|\mathcal{Q}] = 0$, so the covariance is zero. The same argument applies to the second signal-noise term.

Noise-noise term (third term). Since $\mathbb{E}[\nu_t^j|\mathcal{Q}] = \mathbb{E}[\nu_s^k|\mathcal{Q}] = 0$ (the first part of Assumption 1), the covariance equals

$$\mathbb{E}[\nu_t^j\nu_s^k|\mathcal{Q}] = \mathbb{E}[\tilde{\epsilon}_{ijt}\tilde{\epsilon}_{iks}|\mathcal{Q}] - \mathbb{E}[\tilde{\epsilon}_{ijt}\tilde{\epsilon}_{i'ks}|\mathcal{Q}] - \mathbb{E}[\tilde{\epsilon}_{i'jt}\tilde{\epsilon}_{iks}|\mathcal{Q}] + \mathbb{E}[\tilde{\epsilon}_{i'jt}\tilde{\epsilon}_{i'ks}|\mathcal{Q}]. \tag{A6}$$

The second and third terms involve two workers ($i \neq i'$) in different periods ($t \neq s$) and are zero by Assumption 3(ii) and Fact 1. If $j \neq k$, the first and fourth terms also involve distinct matches (the same worker at different firms) at $t \neq s$ and are zero for the same reason. If $j = k$, then they are within-match serial covariances at $t \neq s$ and are zero by Assumption 3(i) and Fact 1. □

Lemma A2 (Restrictions on κ and ω)**.** *Suppose workers are ordered at random within pairs.*

(i) If the second part of Assumption 1 holds, then $\kappa_j = 0$ for every j.

(ii) If Assumption 4 holds, then $\omega_{jk} = 0$ for every $j \neq k$.

(iii) If the second part of Assumption 1 and Assumption 4 hold, then $\bar{\mu}_j = 0$, so that $\tilde{\mu}_{ij} = \mu_{ij}$, and $\omega_{jj} = 2\,\mathrm{Var}(\tilde{\mu}_{ij}|\mathcal{Q})$.

Proof. (i) The second part of Assumption 1 gives $\mathbb{E}[\Delta\mu^j|\mathcal{G}, \alpha, \psi, X] = 0$. Since $\Delta\alpha$ is a function of α, by Fact 1, $\mathbb{E}[\Delta\mu^j|\mathcal{Q}] = 0$ and $\mathbb{E}[\Delta\alpha\Delta\mu^j|\mathcal{Q}] = 0$, so $\kappa_j = 0$.

(ii) By Fact 2, $\mathbb{E}[\Delta\mu^j|\mathcal{Q}] = \mathbb{E}[\Delta\mu^k|\mathcal{Q}] = 0$, so $\omega_{jk} = \mathbb{E}[\Delta\mu^j\Delta\mu^k|\mathcal{Q}]$, which expands into

$$\mathbb{E}[\mu_{ij}\mu_{ik}|\mathcal{Q}] - \mathbb{E}[\mu_{ij}\mu_{i'k}|\mathcal{Q}] - \mathbb{E}[\mu_{i'j}\mu_{ik}|\mathcal{Q}] + \mathbb{E}[\mu_{i'j}\mu_{i'k}|\mathcal{Q}]. \tag{A7}$$

Because $j \neq k$, each product involves two distinct worker-firm matches, so each term is zero by Assumption 4 and Fact 1.

(iii) The event that worker i is at firm j and belongs to a pair in $\mathcal{Q}$ is a function of $\mathcal{G}$, so by the

second part of Assumption 1 and Fact 1, $\bar{\mu}_j = \mathbb{E}[\mu_{ij}|j, \mathcal{Q}] = 0$; hence, $\tilde{\mu}_{ij} = \mu_{ij}$. Expanding,

$$\omega_{jj} = \mathrm{Var}(\mu_{ij}|\mathcal{Q}) + \mathrm{Var}(\mu_{i'j}|\mathcal{Q}) - 2\,\mathrm{Cov}(\mu_{ij}, \mu_{i'j}|\mathcal{Q}). \tag{A8}$$

The two variances are equal by Fact 2. The covariance is zero: $\mathbb{E}[\mu_{ij}|\mathcal{Q}] = \mathbb{E}[\mu_{i'j}|\mathcal{Q}] = 0$ by Fact 1, and $\mathbb{E}[\mu_{ij}\mu_{i'j}|\mathcal{Q}] = 0$ by Assumption 4 and Fact 1, since (i, j) and (i', j) are distinct matches. Hence, $\omega_{jj} = 2\,\mathrm{Var}(\mu_{ij}|\mathcal{Q}) = 2\,\mathrm{Var}(\tilde{\mu}_{ij}|\mathcal{Q})$. □

A.3 Proofs of results in the main text

Proof of Result 1: Under Assumption 2, $\mu_{ij} = 0$ for all (i, j), so $\Delta\mu^j = 0$ and $\kappa_j = \omega_{jk} = 0$ for all j, k. Because $\mu = 0$, Assumption 2 implies Assumption 1. Applying Lemma A1 to the numerator and the denominator,

$$\mathrm{plim}\,\hat{\beta}_1 = \frac{\mathrm{Cov}(Z^B_{t'}, Z^A_{t-1}|\mathcal{P})}{\mathrm{Cov}(Z^A_t, Z^A_{t-1}|\mathcal{P})} = \frac{K_{AB}}{K_{AA}} = \frac{\mathrm{Var}(\Delta\alpha|\mathcal{P})}{\mathrm{Var}(\Delta\alpha|\mathcal{P})} = 1, \tag{A9}$$

where the denominator is strictly positive by the relevance condition.

Proof of Result 2: Apply Lemma A1 as in the proof of Result 1 and then Lemma A2,

$$\mathrm{plim}\,\hat{\beta}_1 = \frac{K_{AB}}{K_{AA}} = \frac{\mathrm{Var}(\Delta\alpha|\mathcal{P}) + \kappa_A + \kappa_B + \omega_{AB}}{\mathrm{Var}(\Delta\alpha|\mathcal{P}) + 2\kappa_A + \omega_{AA}} = \frac{\mathrm{Var}(\Delta\alpha|\mathcal{P})}{\mathrm{Var}(\Delta\alpha|\mathcal{P}) + 2\,\mathrm{Var}(\tilde{\mu}_{ij}|\mathcal{P})}, \tag{A10}$$

where the first equality uses Lemma A1, the second equality is a definition, and the third equality uses Lemma A2. The denominator is strictly positive by the relevance condition.

Proof of Result 3: The sample autocovariance converges to $\mathrm{Cov}(Z^A_t, Z^A_{t-1}|\mathcal{P}) = K_{AA}$, the denominator in the proof of Result 2, which equals $\mathrm{Var}(\Delta\alpha|\mathcal{P}) + 2\,\mathrm{Var}(\tilde{\mu}_{ij}|\mathcal{P})$ by the same argument.

Proof of Result 5: The two firms in each covariance are distinct, so Lemma A1 applies with $\mathcal{Q} = \mathcal{P}_2$ and

$$\mathrm{plim}\,\hat{\beta}_1^{double} = \frac{\mathrm{Cov}(Z^B_{t'}, Z^C_{t''}|\mathcal{P}_2)}{\mathrm{Cov}(Z^A_t, Z^C_{t''}|\mathcal{P}_2)} = \frac{K_{BC}}{K_{AC}} = \frac{\mathrm{Var}(\Delta\alpha|\mathcal{P}_2)}{\mathrm{Var}(\Delta\alpha|\mathcal{P}_2)} = 1, \tag{A11}$$

where the third equality uses Lemma A2(i) and (ii), and the denominator is strictly positive by the relevance condition.

B More details on a direct test for exogenous mobility using predetermined characteristics

This appendix provides additional detail on the claims in Section 1.4. Specifically, it develops the notation to prove a more formal version of Result 4.

B.1 Firm-specific gender gaps are a component of the match effect

Let $f_i \in \{0, 1\}$ indicate whether worker i is female and collect the indicators in $f \equiv (f_1, ..., f_N)$. Because gender is time-invariant, in equation (1) it is absorbed into the person effect. Because we want to be agnostic about how firms price gender, we generalize the wage equation (1) to allow the gender gap to vary across firms:

$$y_{it} = \alpha_i^r + \psi_{j(i,t)} + \gamma_{j(i,t)} f_i + \mu_{ij(i,t)} + \tilde{\epsilon}_{ij(i,t)t}, \tag{A12}$$

where α_i^r is the residual person effect and γ_j is the firm-specific gender gap.

Let $\bar{\gamma}$ denote a constant. This constant is the average gender gap, which is defined using an Assumption below. Define the firm-specific deviations from the average, $g_j \equiv \gamma_j - \bar{\gamma}$. For expositional simplicity, we will often refer to g_j as the firm-specific gender gap. Using these definitions and collecting terms, we can write (A12) as

$$y_{it} = \alpha_i^* + \psi_{j(i,t)} + \mu^*_{ij(i,t)} + \tilde{\epsilon}_{ij(i,t),t}, \qquad \alpha_i^* \equiv \alpha_i^r + \bar{\gamma} f_i, \qquad \mu^*_{ij} \equiv \mu_{ij} + g_j f_i. \tag{A13}$$

This is equation (1) with the person and match effects written with stars. Writing it this way shows, first, that the firm-specific returns term, $g_j f_i$, is part of the match effect because it varies at the worker-firm level, and second, that the assumptions made in the body of the paper apply here when they are stated in terms of (α^*, μ^*).

Because person effects are now two dimensional, we replace α in every conditioning set with the pair (f, α^r) where $\alpha^r \equiv (\alpha_1^r, ..., \alpha_N^r)$. Since α^* is a function of (f, α^r), conditioning on the pair implies conditioning on α^*. So the assumptions below imply those stated in the main text. Fact 1 applies unchanged to this larger conditioning set.

B.2 Exogenous mobility in this notation

We rewrite Assumption 1 with the new structure of match effects.

Assumption A1. *Assume that:*

$$\mathbb{E}[\tilde{\epsilon}_{ijt} | \mathcal{G}, f, \alpha^r, \psi, X, \mu^*] = 0, \qquad \forall i, j, t. \tag{A14}$$

and

$$\mathbb{E}[\mu_{ij}^* | \mathcal{G}, f, \alpha^r, \psi, X] = 0, \qquad \forall i, j. \tag{A15}$$

To see what this assumption implies about the firm-specific gender gaps, g_j, note that f_i is in the conditioning set, so

$$\mathbb{E}[\mu_{ij}^* | \mathcal{G}, f, \alpha^r, \psi, X] = \mathbb{E}[\mu_{ij} | \mathcal{G}, f, \alpha^r, \psi, X] + f_i \mathbb{E}[g_j | \mathcal{G}, f, \alpha^r, \psi, X]. \tag{A16}$$

Assumption A1 only restricts the sum of the two terms on the right-hand side. This assumption does not by itself imply that the firm-specific gender gaps are conditionally mean zero because there are knife-edge cases where this implication does not hold: every woman's residual match effect exactly offsets the firm's deviation, $\mathbb{E}[\mu_{ij}|\cdot] = -\mathbb{E}[g_j|\cdot]$. The following assumption rules out this knife-edge case.

Assumption A2. *Assume that*

(i) $\mathbb{E}[\mu_{ij} | \mathcal{G}, f, \alpha^r, \psi, X] = 0$ *for all* i, j*;*

(ii) $Pr(f_i = 1$ *for some* $i) = 1$*.*

Part (i) is Assumption 1 applied to the "residual" part of the match effect after taking out the firm-specific gender gap. Part (ii) is a non-degeneracy condition that there is at least one woman. Otherwise, there would be no restrictions on the g_j. These two assumptions imply a conditional mean-zero restriction on the firm-specific gender gaps.

Lemma A3. *Suppose Assumption A2 holds. Then the second part of Assumption A1 holds if and only if*

$$\mathbb{E}[g_j | \mathcal{G}, f, \alpha^r, \psi, X] = 0, \qquad \forall j. \tag{A17}$$

Proof. By part (i) of Assumption A2, the decomposition above reduces to $\mathbb{E}[\mu_{ij}^* | \mathcal{G}, f, \alpha^r, \psi, X] = f_i \mathbb{E}[g_j | \mathcal{G}, f, \alpha^r, \psi, X]$. If equation (A17) holds, then the right-hand side is zero for every (i, j), so the second part of Assumption A1 holds.

Going the other way, suppose the second part of Assumption A1 holds. Then $f_i \mathbb{E}[g_j|\cdot] = 0$ almost surely. Fix j and a realization of the conditioning variables. By Part (ii) of Assumption A2, almost surely there exists a woman i with $f_i = 1$, and for that worker the product can be zero only if $\mathbb{E}[g_j|\cdot] = 0$ at that realization. Since this holds almost surely, condition (A17) holds. □

Equation (A17) implies that $\bar{\gamma}$ is the common gender gap. It also implies that given the conditioning set, the firm-specific gender gaps are mean zero. Heuristically, workers do not select firms on the basis of the g_j.

B.3 What the gender instrument identifies

We can now write the within-firm gaps between the paired movers:

$$Z^j = \Delta\alpha^r + \gamma_j \Delta f + \Delta\mu^j + \nu^j, \qquad j \in \{A, B\}, \tag{A18}$$

where $\Delta f \equiv f_i - f_{i'} \in \{-1, 0, 1\}$, $\Delta\alpha^r \equiv \alpha^r_i - \alpha^r_{i'}$, and $\Delta\mu^j$ and ν^j are as defined in Appendix A.

The instrument. As in the text, let $c_{ii'} \in \mathcal{C} \equiv \{MM, FF, MF, FM\}$ denote the gender configuration of the pair. Let $D^c_{ii'} \equiv \mathbf{1}\{c_{ii'} = c\}$ so that

$$D^{MM}_{ii'} = (1 - f_i)(1 - f_{i'}), \qquad D^{FF}_{ii'} = f_i f_{i'}, \qquad D^{MF}_{ii'} = (1 - f_i) f_{i'}, \qquad D^{FM}_{ii'} = f_i (1 - f_{i'}).$$

The four dummies sum to one. The instrument set is $D_{ii'} \equiv (D^{FF}_{ii'}, D^{MF}_{ii'}, D^{FM}_{ii'})'$, with MM as the omitted category. Each dummy is a function of f, and so is in the conditioning sets of Assumptions A1 and A2.

The TSLS estimand for β_1 in equation (2) with instruments $D_{ii'}$ is

$$\operatorname{plim} \hat{\beta}_1^{gender} = \frac{\operatorname{Cov}(Z^A, D_{ii'}|\mathcal{P})' \operatorname{Var}(D_{ii'}|\mathcal{P})^{-1} \operatorname{Cov}(D_{ii'}, Z^B|\mathcal{P})}{\operatorname{Cov}(Z^A, D_{ii'}|\mathcal{P})' \operatorname{Var}(D_{ii'}|\mathcal{P})^{-1} \operatorname{Cov}(D_{ii'}, Z^A|\mathcal{P})}, \tag{A19}$$

where $\operatorname{Cov}(Z^j, D_{ii'}|\mathcal{P})$ is the 3×1 vector with elements $\operatorname{Cov}(Z^j, D^c_{ii'}|\mathcal{P})$ for $c \in \{FF, MF, FM\}$. When the instrument is scalar it reduces to the standard covariance ratio.

Result A1. *Consider estimating equation (2) by two-stage least squares, where $Z^A_{ii't}$ is instrumented with $D_{ii'}$ in the model with firm-specific gender gaps so that the gaps are given by equation (A18). Suppose the following relevance condition holds*

$$\operatorname{Var}(D_{ii'}|\mathcal{P}) \text{ is nonsingular and } \operatorname{Cov}(Z^A, D_{ii'}|\mathcal{P}) \neq 0 \tag{A20}$$

and that Assumptions A1 and A2 hold so that condition (A17) holds. Then,

$$\operatorname{Cov}(Z^B, D_{ii'}|\mathcal{P}) = \operatorname{Cov}(Z^A, D_{ii'}|\mathcal{P}), \tag{A21}$$

and hence

$$\operatorname{plim} \hat{\beta}_1^{gender} = 1. \tag{A22}$$

The first part of the relevance condition says that each gender configuration occurs with positive probability among the paired movers. The second part says that the instruments are correlated with the origin gap.

Proof. Fact 1 applies with the conditioning set $(\mathcal{G}, f, \alpha^r, \psi, X)$, or with μ^* added: if V has conditional mean zero given that set, then $\operatorname{Cov}(V, D^c_{ii'}|\mathcal{P}) = 0$ for every c because $D^c_{ii'}$ is a function of

f.

Fix c and $j \in \{A, B\}$. Substituting $\gamma_j = \bar{\gamma} + g_j$ into (A18),

$$\begin{aligned}\operatorname{Cov}(Z^j, D^c|\mathcal{P}) = \operatorname{Cov}(\Delta\alpha^r, D^c|\mathcal{P}) + \bar{\gamma}\operatorname{Cov}(\Delta f, D^c|\mathcal{P}) \\ + \operatorname{Cov}(g_j\Delta f, D^c|\mathcal{P}) + \operatorname{Cov}(\Delta\mu^j, D^c|\mathcal{P}) + \operatorname{Cov}(\nu^j, D^c|\mathcal{P}). \qquad \text{(A23)}\end{aligned}$$

We want to show that the three terms on the second line are zero. For the first term: because Δf is a function of f, $\mathbb{E}[g_j\Delta f|\mathcal{G}, f, \alpha^r, \psi, X] = \Delta f\mathbb{E}[g_j|\mathcal{G}, f, \alpha^r, \psi, X] = 0$ by (A17). For the second term: $\mathbb{E}[\Delta\mu^j|\mathcal{G}, f, \alpha^r, \psi, X] = 0$ by Assumption A2(i). For the third term: $\mathbb{E}[\nu^j|\mathcal{G}, f, \alpha^r, \psi, X, \mu^*] = 0$ by the first part of Assumption A1.

Now all that remains is the first line. These terms do not depend on j. Hence, $\operatorname{Cov}(Z^j, D^c|\mathcal{P})$ is the same for $j = A$ and $j = B$. Stacking over c gives (A21). Substituting into (A19) gives a ratio of identical quantities, which is 1. □

This result shows that instrumenting with a predetermined characteristic is a test of exogenous mobility (and mild regularity conditions). Heuristically, if the gender gap is systematically different in origin and destination firms then this pattern generates a coefficient that departs from 1 and reflects workers systematically choosing firms on the basis of the gender-specific pay premium.

The same logic developed in this Appendix applies to any predetermined characteristic. For a scalar characteristic x_i such as years of education or age, replace f_i by x_i, let γ_j be the firm-specific return to the characteristic, replace Part (ii) of Assumption A2 by $\Pr(x_i \neq 0) > 0$ for some i, and use the gap $\Delta x_{ii'} \equiv x_i - x_{i'}$ as a single instrument. The proof above goes through with D^c replaced by Δx, and the relevance condition becomes $\operatorname{Cov}(Z^A, \Delta x|\mathcal{P}) = \mathbb{E}[\Delta\alpha^r\Delta x|\mathcal{P}] + \bar{\gamma}\operatorname{Var}(\Delta x|\mathcal{P}) \neq 0$, where the equality uses $\mathbb{E}[\Delta x|\mathcal{P}] = 0$ by Fact 2.

C Forward, backward, and donut paired movers

In the baseline implementation, we regress the gap in earnings at the destination firm at $t = 1$ (first year at the destination) on the gap in earnings at the origin firm at $t = 0$ (last year at the origin), instrumenting with the lagged gap in earnings at the origin firm at $t = -1$. However, we can use any three periods in which we observe two workers in two firms. These three periods do not need to be consecutive, and this regression can be implemented in either direction (regressing the gap at the destination on the gap at the origin, or vice versa).

Panel A of Appendix Table A15 shows results from the baseline, or forward, version of the regression, where the outcome is the gap in the destination firm. Column 1 replicates the baseline results from Column 1 of Table 4—where the outcome is the gap in earnings at the destination firm at $t = 1$ and the endogenous variable is the gap in earnings at the origin firm at $t = 0$, instrumented with the lagged gap in earnings at the origin firm at $t = -1$.

Column 2 shows results when we keep the outcome the same but change the right-hand side variable. Instead of the last year at the origin firm ($t = 0$), we use the gap in earnings at the

origin firm at $t = -1$ (second to last period at the origin firm) instrumented with the lagged gap in earnings at the origin firm at $t = -2$ (third to last period at the origin firm). Column 3 shows results when we exclude the first year at the destination ($t = 1$) from the sample, using as the outcome the gap in earnings at the destination firm in the second year, $t = 2$, and as the regressor the gap in earnings at the origin firm at $t = 0$, instrumented with the lagged gap in earnings at the origin firm at $t = -1$.

Finally, Column 4 shows results from a donut version of the test, combining Columns 2 and 3 so that we omit the last year at the origin firm and the first year at the destination firm. Columns 5 to 8 replicate Columns 1 to 4 within a fixed sample of movers observed in all periods. The results are very similar across all specifications, with coefficients ranging from 0.75 to 0.77.

Panel B of Appendix Table A15 shows results from the backward version of the paired-mover coefficient, where the outcome is the earnings gap in the origin firm, and the independent variable is the earnings gap at the destination. Across specifications, the coefficient is larger than the baseline when the right-hand side variable is the gap in earnings in the first year at the destination—around 0.85—and closer to the baseline when the right-hand side variable is the gap in earnings in the second year at the destination—around 0.75. This pattern is consistent with earnings in the first year at the destination being noisier, which can also be seen in Table A5. Overall, the results suggest that our findings are not driven by the specific way we implement the paired-mover regression.

D Paired movers based on larger and smaller gaps

This estimator relies on the earnings gap between the paired movers. Panel (a) of Figure A3 shows the CDF of the absolute gap, which is closely related to the variance. While the median is around 0.25, there is a wide support. For the remainder of the figure we divide the paired movers into deciles of the absolute earnings gap based on the instrument (the lagged earnings gap).

The paired-mover coefficient could be driven by the instability of small earnings gaps at the origin firm. Panel (b) shows instead that the coefficient is *declining* in the size of the gap at the origin firm. The coefficient is smallest where the gap is largest.

The earnings gaps might also be less stable when they are larger. By decile, we compute the implied AR(1) coefficient on earnings gaps. Panel (c) shows that apart from the first two deciles, the gaps are *more* stable the larger they are.

There are corners of the data with a very large role for match effects. Panel (d) computes the variance of match effects under exogenous mobility (see Appendix Table A16 for a tabular version).[11] In the top decile the variance of match effects is about 0.20, which is more than 6 times the overall number. Summing the five estimates and dividing by ten, the implied variance of match effects is 0.028, slightly larger than the pooled estimate.

[11] The panel only shows this calculation for the top 5 deciles, because these are the deciles where the paired-mover coefficient is less than 1, which is required by the formula.

E Heuristic identification

Appendix Figure A4 shows how the two parameters most closely tied to match effects, σ_m^2 (variance of match effects) and σ_h^2 (variance of person worker effects), map into the moments that identify them. Each panel starts from the baseline estimates (Table 7) and varies one parameter, holding the others fixed.

In the baseline calibration, the paired-mover coefficient is decreasing in σ_m^2 (panel (a)) and increasing in σ_h^2 (panel (d)). Result 2 gives the intuition under exogenous mobility: the coefficient depends on the variance of match effects relative to the variance of the person-effect gap. When person effects are large relative to match effects, they dominate the gap, so the gap is stable across firms. The coefficient therefore pins down the relative size of the two variances, and the autocovariance of the origin gap, which rises with both, pins down their level.

The alternative calibration (Table A10) instead targets the three cross-firm covariances of the gap among paired double movers. A larger σ_h^2 raises all three in parallel (panel (e)) and leaves the double-mover coefficients roughly unchanged (panel (f)). A larger σ_m^2 also raises them, but by different amounts (panel (b)), so the double-mover coefficients fall (panel (c)). The ratios of the covariances therefore identify σ_m^2, and their level then identifies σ_h^2.

Table A1: Distribution of # of workers per origin-destination pair

	Min (1)	p5 (2)	p10 (3)	p25 (4)	p50 (5)	p75 (6)	p90 (7)	p95 (8)	Max (9)
All OD pairs	1	2	2	2	4	10	37	83	1,507
OD pairs in paired-mover sample	2	2	2	2	4	11	38	86	1,507

This table shows the distribution of the number of eligible workers per origin-destination-year pair, weighted by the number of workers in each pair. Eligible workers are those that are observed for at least two years at the origin firm. We show the distribution of the number of eligible workers for all possible origin-destination-year pairs (first row) and among pairs in the paired-mover sample (second row).

Table A2: Paired-mover coefficient, robustness to residualization approach

	Residualization		Matching by age and gender	
	Baseline (CCHK) (1)	Without education (2)	1-year age bin (3)	10-year age bin (4)
A. Brazil, Second Stage				
Gap in Origin	0.758	0.750	0.742	0.700
	(0.0005)	(0.0005)	(0.001)	(0.0009)
Observations	3,386,479	3,386,479	549,834	1,408,531
B. Brazil, 1st Stage				
Gap in Origin (lag)	0.918	0.916	0.920	0.915
	(0.0004)	(0.0004)	(0.001)	(0.0007)
F-statistic	6,473,532.2	6,154,940.6	822,427.0	1,925,253.4

This table shows results from a robustness exercise in which we vary the residualization/matching approach used to construct the earnings gaps. Column 1 shows the baseline results, where we residualize log earnings on a third-degree polynomial in (normalized) age interacted with gender and four education categories, omitting the linear term in age. Column 2 shows the results when we do not include education. Columns 3 and 4 show results when we match paired movers on age and gender. In Column 3 we use one-year age bins, while in Column 4 we use ten-year age bins.

Table A3: Paired-mover coefficient, reweighting for the firm size distribution

		Reweighted ... to match overall firm size distribution	
	Baseline	Origin	Destination
	(1)	(2)	(3)
A. Brazil, Second Stage			
Gap in Origin	0.758	0.803	0.686
	(0.0005)	(0.0006)	(0.0006)
Observations	3,386,479	3,386,479	3,386,479
B. Brazil, 1st Stage			
Gap in Origin (lag)	0.918	0.853	0.842
	(0.0004)	(0.0004)	(0.0004)
F-statistic	6,473,532.2	4,888,359.4	4,434,552.3

This table shows results from a robustness exercise in which we reweight the sample of paired movers to match the overall firm size distribution in the data. We first divide the firm size distribution in the full data (Column 1, Table 1) into 100 bins with the same number of firms. We then reweight the sample of paired movers so that the distribution of firm sizes at origin (Column 2) and destination firms (Column 3) matches the overall distribution in the data, within the shared support of firm sizes in the paired-mover sample. In practice, the first 27 bins of the firm size distribution are not in the support of the paired-mover sample, and more than 95% of the firms in the paired-mover sample are in the top 10% of the firm size distribution (larger than 15 employees).

Table A4: Earnings gap AR(1) coefficients

	Full sample	Destination	Origin
	(1)	(2)	(3)
A. Balanced Panel			
θ	-0.050	-0.060	-0.043
	(0.006)	(0.004)	(0.001)
AR(1) Coefficient	0.951	0.942	0.958
B. Full Sample			
θ	-0.038	-0.038	-0.038
	(0.006)	(0.001)	(0.002)
AR(1) Coefficient	0.963	0.963	0.962
Observations	24	9	14

This table shows coefficients from regression $\ln(\rho_k) = \theta k + \eta_k$, where ρ_k is the autocorrelation of the log earnings gap between paired movers at time lag/lead k. Hence, $\exp(\theta)$ is the AR(1) coefficient of the log earnings gap. The autocorrelations are the same as those shown in Table A5. Panel A shows the results for the balanced panel, while Panel B shows the results for the full sample. Column 1 shows the results using all available autocorrelations, while Columns 2 and 3 show the results using only the autocorrelations at destination and origin firms. Regressions are weighted by the number of observations used to calculate each autocorrelation. Standard errors are in parentheses.

Table A5: Earnings gap autocorrelation and implied AR(1) coefficients

	Balanced Panel					Full Sample				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
A. Origin firm, autocor.										
Time lag	-1	-2	-3	-4	-5	-1	-2	-3	-4	-5
0	0.852	0.817	0.785	0.756	0.720	0.823	0.786	0.763	0.743	0.713
-1		0.942	0.903	0.867	0.827		0.927	0.884	0.853	0.817
-2			0.940	0.900	0.856			0.932	0.892	0.851
-3				0.939	0.890				0.933	0.886
-4					0.932					0.929
B. Origin firm, AR(1) coef.										
Time lag	-1	-2	-3	-4	-5	-1	-2	-3	-4	-5
0		0.959	0.961	0.963	0.952		0.955	0.971	0.974	0.960
-1			0.959	0.960	0.954			0.954	0.965	0.958
-2				0.957	0.951				0.957	0.954
-3					0.948					0.950
C. Destination firm, autocor.										
Time lead		2	3	4	5		2	3	4	5
1		0.936	0.891	0.851	0.788		0.867	0.833	0.803	0.777
2			0.938	0.893	0.827			0.886	0.849	0.818
3				0.936	0.864				0.895	0.859
4					0.903					0.901
D. Destination firm, AR(1) coef.										
Time lead		2	3	4	5		2	3	4	5
1			0.952	0.955	0.926			0.961	0.964	0.968
2				0.952	0.926				0.958	0.963
3					0.923					0.960

This table shows the autocorrelation and implied AR(1) coefficients of the log earnings gap between paired movers at origin and destination firms for different time lags/leads. Panels A and C show the autocorrelation of the log earnings gap at origin and destination firms, respectively. Hence, the first row of Column 1 in Panel A shows the correlation between the log earnings gap at the last year at the origin firm ($t = 0$) and one year before that ($t = -1$). Panels B and D show the implied AR(1) coefficients at origin and destination firms, respectively, calculated as the ratio of the autocorrelation at time lag/lead k and $k + 1/k - 1$. We show results for both a balanced panel where all paired movers stay at the origin firm from $t = -5$ to $t = 0$ and at the destination firm from $t = 1$ to $t = 5$ (Columns 1 to 5) and the full sample of paired movers observed in a given time lag/lead (Columns 6 to 10).

Table A6: The paired-mover coefficient in Brazilian data: additional splits

	Sector		Firm Pay			
	Stayers (1)	Switchers (2)	Low-Low (3)	Low-High (4)	High-High (5)	High-Low (6)
A. 2nd Stage						
Gap in Origin	0.779	0.735	0.767	0.748	0.770	0.683
	(0.0007)	(0.0007)	(0.0009)	(0.002)	(0.0008)	(0.001)
p-value, equality test	0.0000		0.0000		0.0000	
$\text{Cov}(Z_t, Z_{t-1})$	0.239	0.203	0.140	0.178	0.322	0.199
$\text{Var}(\tilde{\mu}_{ij} \mid \mathcal{P})$	0.026	0.027	0.016	0.022	0.037	0.032
Observations	1,617,455	1,769,024	1,168,363	407,561	1,177,187	399,110
B. 1st Stage						
Gap in Origin (lag)	0.919	0.917	0.908	0.907	0.929	0.913
	(0.0005)	(0.0005)	(0.0006)	(0.001)	(0.0006)	(0.001)
F-statistic	3,334,933.1	3,134,490.1	1,979,006.2	725,164.4	2,482,883.8	622,908.5

This table shows the paired-mover coefficient in different splits of the sample in Brazil. Columns 1 and 2 split paired movers by whether they stay in the same sector or move to a different sector (2-digit CNAE code). Columns 3 to 6 split paired movers by moves across low/high pay firms. We define low/high pay as the bottom/top 50% of the firm pay distribution among firms in the sample. Firm pay is computed yearly across all workers, not only paired movers.

Table A7: Summary statistics: paired movers

	Sector		Firm Pay			
	Stayers (1)	Switchers (2)	Low-Low (3)	Low-High (4)	High-High (5)	High-Low (6)
Avg. age	33.83	33.04	33.52	32.40	33.39	33.86
Share female	0.33	0.30	0.41	0.32	0.23	0.29
Years of schooling	10.79	10.89	10.16	10.89	11.64	10.50
Avg. Tenure (months)	32.80	31.19	29.85	29.02	34.67	34.57
Median Tenure (months)	21.18	20.31	20.48	19.32	21.82	22.22
Avg. log earnings	2.43	2.38	2.00	2.25	2.88	2.35
Predicted log earnings (firm size)	2.47	2.48	2.46	2.47	2.51	2.48
# of Worker-Years	3,234,910	3,538,048	2,336,726	815,122	2,354,374	798,220
# of Workers	2,955,547	3,228,003	2,151,148	805,167	2,102,149	785,147
Avg. (origin) firm pay	2.46	2.47	2.01	2.29	2.83	2.50
Avg. (origin) firm size	242.07	274.99	178.41	398.54	328.13	426.88
# of (origin) Firm-Years	570,546	569,459	419,619	163,745	297,308	145,240
# of (origin) Firms	289,981	277,655	234,970	102,681	125,417	84,029
Var(Z)	0.327	0.291	0.202	0.252	0.441	0.299

This table shows descriptive statistics of different splits of the paired-mover sample in Brazil. Columns 1 and 2 display movers split by whether they stay in the same sector or move to a different sector (2-digit CNAE code). Columns 3 to 6 display movers across different pay bins. We define low/high pay as the bottom/top 50% of the firm pay distribution among firms in the sample. Firm pay is computed yearly across all workers, not only paired movers.

Table A8: Within-sector moves: observed vs. random mobility

	Share of moves within-sector		
	Observed (1)	Random (Uniform) (2)	Random (Weighted) (3)
All movers	0.381	0.011	0.061
Paired movers	0.478	0.011	0.039

This table shows the observed fraction of moves that are within-sector (Column 1), compared to the fraction of moves that would be within-sector under two random mobility counterfactuals. In Column 2, the counterfactual is a uniform distribution over all possible sectors. With 88 sectors in the data, $1/88 = 1.1\%$ of moves would be within-sector. In Column 3, the counterfactual is a weighted distribution over all possible sectors, where the weights are given by sector employment shares. In this case, the probability of a within-sector move for a given worker is the employment share of her sector, and the overall probability of a within-sector move is the average across all workers.

Table A9: Paired-mover coefficient by earnings of the first mover

	Gap in destination			
	All paired movers		Gap higher than 0.2 log points	
Higher earner first	No	Yes	No	Yes
	(1)	(2)	(3)	(4)
A. Second stage				
Gap in Origin	0.746	0.769	0.725	0.754
	(0.0008)	(0.0008)	(0.0008)	(0.0008)
$\text{Cov}(Z_t, Z_{t-1})$	0.182	0.273	0.332	0.436
$\text{Var}(\tilde{\mu}_{ij} \mid \mathcal{P})$	0.023	0.032	0.046	0.054
Observations	1,978,102	1,408,377	1,062,838	871,790
B. First Stage				
Gap in Origin (lag)	0.898	0.938	0.956	0.983
	(0.0005)	(0.0006)	(0.0006)	(0.0007)
F-statistic	3,769,360.7	2,706,906.7	2,436,622.6	1,921,541.0

This table shows the paired-mover coefficient split by whether the higher earner in the origin firm is the first or second mover. Columns 1 and 2 show the results for all paired movers, while Columns 3 and 4 show the results for paired movers where the gap in earnings is higher than 20 log points. Columns 1 and 3 show the results for pairs where the higher earner in the origin firm is the second mover, while Columns 2 and 4 show the results for pairs where the higher earner in the origin firm is the first mover. We also report components of the exogenous mobility estimator for the variance of match effects following Equation 3.

Table A10: Model results, double-mover calibration

	(1)	(2)	(3)	(4)	(5)	(6)
A. Model fit						
	EE rate	β_1	Cov(Z_t,Z_{t-1})	Var(ψ_j)	Var(α_i)	Cov(α_i, ψ_j)
Empirical Moments (Data)	0.038	0.758	0.220	0.065	0.294	0.047
Simulated Moments (Model)	0.038	0.574	0.313	0.065	0.178	0.048
B. Model parameters						
	λ_0 (UE rate)	δ (EU rate)				
Calibrated Parameters	0.072	0.076				
	σ_m^2	σ_h^2	σ_p^2	a	b	
Estimated Parameters	0.075	0.068	0.068	0.132	1.540	
C. Structural variances (steady-state)						
	m_{ij}	h_i	p_j			
Mean	0.174	0.046	0.117			
Variance	0.094	0.075	0.070			
Cov(h_i, p_j)	0.031					
Cov(h_i, m_{ij})	0.042					
Cov(p_j, m_{ij})	0.016					
D. Double-mover moments						
	Cov(Z^A, Z^B)	Cov(Z^B, Z^C)	Cov(Z^A, Z^C)	β_1	β_2	β_3
Data	0.136	0.134	0.124	0.908	0.922	0.985
Model	0.122	0.133	0.130	0.918	0.936	0.981
E. Implied variance of match effects under exogenous mobility						
	Var($\tilde{\mu}_{ij} \mid \mathcal{P}$)					
Data	0.027					
Model	0.067					

This table shows an alternative model calibration where we target double-mover moments rather than paired-mover moments. We fix the unemployed arrival rate λ_0 and the separation rate δ to quarterly UE and EU rates in Brazil, measured using labor force surveys (PNAD). We estimate the five remaining parameters of the model: the on-the-job arrival rate $\lambda_{1i} = \min\{\max\{0, a + b \times h_i\}, 1\}$, the variance of match effects σ_m^2, the variance of person effects σ_h^2, and the variance of firm effects σ_p^2. We match the following moments: the EE rate, the double-mover coefficients, the gap covariance across the three firms of paired double movers Cov(Z_A, Z_B), ..., the variance of firm effects $Var(\psi_j)$, and the covariance of firm and person effects Cov(α_i, ψ_j). Unlike in the baseline calibration, β_1 and Cov(Z_t,Z_{t-1}) are untargeted. We simulate 200,000 workers and 200 firms and pull 100 periods out of the steady-state distribution to compute the moments.

Table A11: Bottom 20% to top 20% moves: accepted and rejected offers

	Mean change in firm FE (1)	Mean change in earnings (2)
A. Baseline calibration		
Accepted offers	0.42	0.39
Rejected offers	0.27	-0.17
B. Double-mover calibration		
Accepted offers	0.46	0.44
Rejected offers	0.34	-0.32

This table shows the mean change in firm effects (Column 1) and the associated mean change in earnings (Column 2) for accepted and rejected offers in the model simulations. We show offers received by workers in the bottom 20% of the firm effect distribution from firms in the top 20% of the firm effect distribution. Panel A shows the results for the baseline calibration in Table 7, while Panel B shows the results for the double-mover calibration in Table A10.

Table A12: Event study test

	Change in $\hat{\psi}_1$ / Change in Log Earnings		
	All movers	Paired movers	Paired double movers
	(1)	(2)	(3)
A. Brazil, Second stage			
Change in $\hat{\psi}_1$	1.09	1.05	0.992
	(0.001)	(0.002)	(0.004)
Observations	4,457,623	730,778	285,657
B. Brazil, First stage			
Change in $\hat{\psi}_2$	0.746	0.831	0.817
	(0.0003)	(0.0006)	(0.001)
F-statistic	5,535,025.1	1,637,287.5	561,436.7
C. Veneto, Second stage			
Change in $\hat{\psi}_1$	1.18	0.663	
	(0.007)	(0.045)	
Observations	281,075	3,829	
D. Veneto, First stage			
Change in $\hat{\psi}_2$	0.500	0.740	
	(0.002)	(0.011)	
F-statistic	95,639.4	4,279.4	

This table shows the event study test in the data. We use a sample of movers to estimate a TSLS regression where the change in the log earnings between destination and origin firms is regressed on the change in the estimated firm effect across origin-destination firms, where we use the firm effects estimated in one sample to instrument the firm effects in the second sample. We estimate two sets of firm fixed effects by randomly splitting the full data (Column 1, Table 1) into two samples. Column 1 shows the results for all movers used in the AKM estimation. Column 2 subsets the sample for movers in the paired-mover sample (Columns 3-4, Table 1). Column 3 subsets the sample for movers in the double-mover sample (Columns 5-7, Table 1). Panel A shows the second stage of the test in Brazilian data, while Panel C shows the second stage in Veneto data. The first stage is shown in Panels B and D.

Table A13: Movers tests in model-simulated data

	Paired mover	Double movers			Event study		
		IV: Firm A	IV: Firm B	IV: Firm C	All	EE	EUE
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
A. Paired-mover calibration							
Gap in origin	0.757						
	(0.0001)						
Gap in C (TSLS)		0.968	0.969				
		(0.008)	(0.008)				
Gap in A (TSLS)				0.998			
				(0.008)			
Change in $\hat{\psi}$					1.00	0.541	1.04
					(0.001)	(0.001)	(0.001)
Observations	23,428,502	17,077	17,077	17,077	946,428	412,247	534,181
Var(Z)	0.215						
Var($\tilde{\mu}_{ij} \mid \mathcal{P}$)	0.026						
Cov(Z_A, Z_B)		0.128	0.128				
Cov(Z_B, Z_C)			0.132	0.132			
Cov(Z_A, Z_C)		0.133		0.133			
B. Double-mover calibration							
Gap in origin	0.574						
	(0.0002)						
Gap in C		0.936	0.918				
		(0.016)	(0.016)				
Gap in B				0.981			
				(0.017)			
Change in $\hat{\psi}$					1.01	0.422	1.08
					(0.001)	(0.001)	(0.001)
Observations	19,581,774	15,255	15,255	15,255	922,876	399,469	523,407
Var(Z)	0.313						
Var($\tilde{\mu}_{ij} \mid \mathcal{P}$)	0.067						
Cov(Z_A, Z_B)		0.122	0.122				
Cov(Z_B, Z_C)			0.133	0.133			
Cov(Z_A, Z_C)		0.130		0.130			

This table shows the paired-mover coefficient (Column 1), the double-mover coefficient (Columns 2-4), and the event study test (Columns 5-7) in data simulated from the baseline model from Table 7 (Panel A) and from the alternative calibration in Table A10 (Panel B). For the paired-mover coefficient, we report the associated variance of match effects following Equation 3. For the double-mover coefficient, we report the specifications with $\beta < 1$ and the associated gap covariances across firms. For the event study test, we estimate firm effects in two randomly split samples of the data and report the results for EE and EUE moves.

Table A14: Simulated Borovičková and Shimer (2024) data

	Paired mover	Paired double movers			Event study		
		IV: Firm A	IV: Firm B	IV: Firm C	All	EE	EUE
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Gap in origin	0.643						
	(0.0001)						
Gap in B (TSLS)		0.986		0.962			
		(0.005)		(0.005)			
Gap in C (TSLS)			0.976				
			(0.005)				
Change in $\hat{\psi}$ (TSLS)					0.942	0.342	0.932
					(0.003)	(0.002)	(0.004)
Observations	27,549,722	99,416	99,416	99,416	1,001,518	533,559	467,959
$\text{Var}(Z)$	0.018						
$\text{Var}(\tilde{\mu}_{ij} \mid \mathcal{P})$	0.003						
$\text{Cov}(Z_A, Z_B)$		0.0089	0.0089				
$\text{Cov}(Z_B, Z_C)$			0.0091	0.0091			
$\text{Cov}(Z_A, Z_C)$		0.0088		0.0088			

This table shows the results using data simulated from the model of Borovičková and Shimer (2024), with the same parameters as estimated in the original paper. We report the paired mover (Column 1), the double mover test (Columns 2-4), and the event study test (Columns 5-7). For the paired-mover coefficient, we report the associated variance of match effects following Equation 3. For the double-mover coefficient, we report the specifications with $\beta < 1$ and the associated gap covariances across firms. For the event study test, we estimate firm effects in two randomly split samples of the data and report the results for EE and EUE moves.

Table A15: Forward and backward paired-mover coefficient

A. "Forward"								
Dependent Var.:	Gap in Destination, $t=1$		Gap in Destination, $t=2$		Gap in Destination, $t=1$		Gap in Destination, $t=2$	
	Baseline	Excl. $t=0$	Excl. $t=1$	"Donut"	Baseline	Excl. $t=0$	Excl. $t=1$	"Donut"
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A1. 2nd Stage								
Gap in Origin, $t=0$	0.758		0.752		0.771		0.756	
	(0.0005)		(0.0007)		(0.0007)		(0.0008)	
Gap in Origin, $t=-1$		0.768		0.761		0.773		0.761
		(0.0006)		(0.0008)		(0.0007)		(0.0008)
Observations	3,386,479	2,145,125	2,381,313	1,530,863	1,530,863	1,530,863	1,530,863	1,530,863
A2. 1st Stage								
Gap in Origin, $t=-1$	0.918		0.924		0.929		0.929	
	(0.0004)		(0.0004)		(0.0005)		(0.0005)	
Gap in Origin, $t=-2$		0.698		0.700		0.700		0.700
		(0.0004)		(0.0004)		(0.0004)		(0.0004)
F-statistic	6,473,532.2	3,706,431.2	4,620,625.5	2,737,149.8	3,337,540.5	2,737,149.8	3,337,540.5	2,737,149.8
B. "Backward"								
Dependent Var.:	Gap in Origin, $t=0$		Gap in Origin, $t=-1$		Gap in Origin, $t=0$		Gap in Origin, $t=-1$	
	Baseline	Excl. $t=0$	Excl. $t=1$	"Donut"	Baseline	Excl. $t=0$	Excl. $t=1$	"Donut"
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
B1. 2nd Stage								
Gap in Destination, $t=1$	0.860		0.849		0.863		0.855	
	(0.0008)		(0.0006)		(0.0009)		(0.0007)	
Gap in Destination, $t=2$		0.755		0.746		0.755		0.746
		(0.0009)		(0.0007)		(0.0009)		(0.0007)
Observations	2,381,313	1,725,617	2,381,313	1,725,617	1,725,617	1,725,617	1,725,617	1,725,617
B2. 1st Stage								
Gap in Destination, $t=2$	0.672		0.672		0.704		0.704	
	(0.0004)		(0.0004)		(0.0004)		(0.0004)	
Gap in Destination, $t=3$		0.699		0.699		0.699		0.699
		(0.0005)		(0.0005)		(0.0005)		(0.0005)
F-statistic	3,034,479.2	2,107,878.4	3,034,479.2	2,107,878.4	2,482,526.9	2,107,878.4	2,482,526.9	2,107,878.4

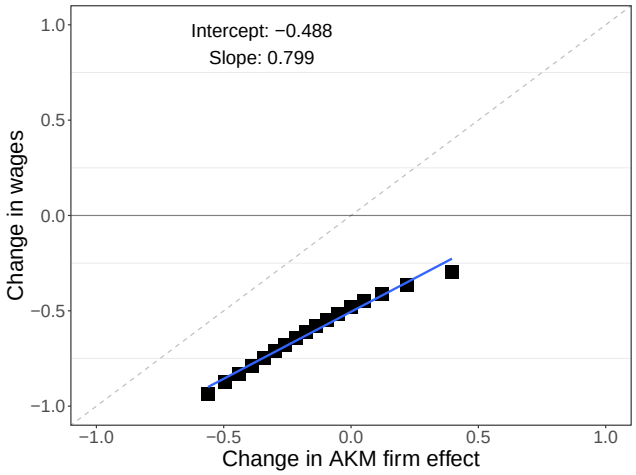
This table shows results from the "forward" (baseline) and "backward" versions of the paired-mover coefficient. Panel A shows results from the forward version, where the outcome is the earnings gap in the destination firm and the right hand side is the earnings gap at the origin firm. Panel B shows results from the backward version, where the outcome is the earnings gap at the origin firm and the right hand side is the earnings gap at the destination firm. In both panels, the first stage results are shown in the lower half and the second stage results are shown in the upper half. Columns 1 and 5 show baseline results using the gap in earnings at $t = 0$ (Panel A) or $t = 1$ (Panel B) as regressor. Columns 2 and 6 exclude the contemporaneous period from the regression, i.e., they use only the lagged gap in earnings at the origin (Panel A) or destination (Panel B) as regressor. Columns 3 and 7 exclude one period after/before the move from the regression, i.e., they use only the gap in earnings at $t = 2$ (Panel A) or $t = -1$ (Panel B) as dependent variable. Finally, Columns 4 and 8 implement a "donut" version of the paired-mover coefficient, where both the contemporaneous period and one period after/before the move are excluded from the regression. Columns 1–4 show results for the full sample of paired movers, while Columns 5–8 show results for a balanced sample of paired movers who are observed in all periods used in the panel.

Table A16: $\text{Var}(\tilde{\mu}_{ij})$ by decile of the absolute earnings gap

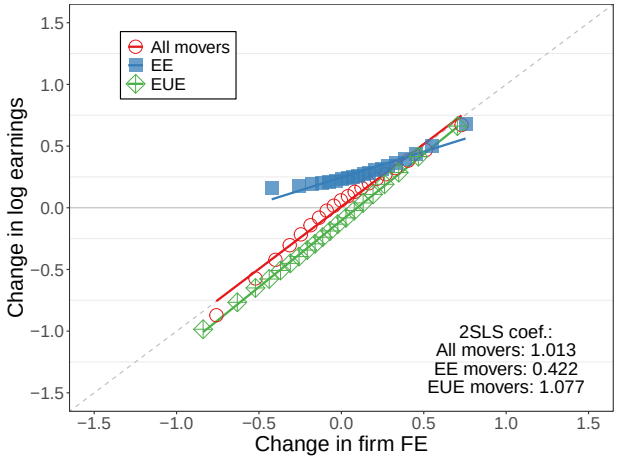
Decile	Mean Gap (1)	β (2)	$\text{Var}(Z)$ (3)	$\text{Cov}(Z_t, Z_{t-1})$ (4)	$\text{Var}(\tilde{\mu}_{ij} \mid \mathcal{P})$ (5)
6	0.287	0.872	0.083	0.068	0.004
7	0.378	0.801	0.144	0.118	0.012
8	0.503	0.761	0.255	0.206	0.025
9	0.710	0.743	0.511	0.399	0.051
10	1.333	0.718	2.004	1.343	0.190

This table shows the underlying numbers for Figure A3. Column 1 shows the mean absolute earnings gap at origin firms for each decile of the gap distribution where $\beta < 1$ (deciles 6 to 10). We define the deciles based on the instrument (lagged earnings gap). Column 2 shows the paired-mover coefficient β estimated separately for each decile. Columns 3 and 4 show the variance and covariance of the earnings gap at the origin firm. Finally, Column 5 shows the variance of match effects implied by Equation (3).

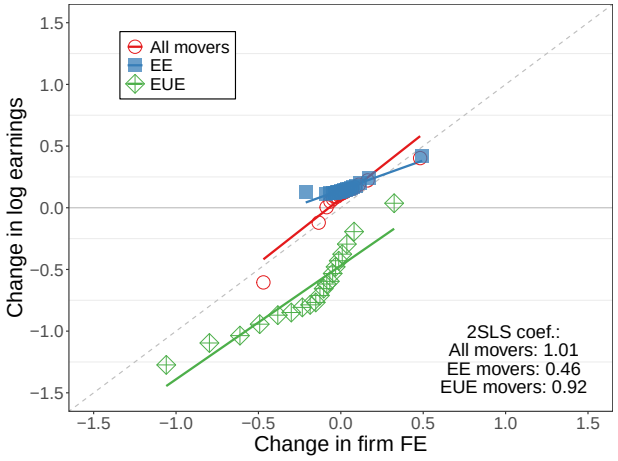
Figure A1: Model simulated data, double-mover calibration



(a) AKM firm effects predictions for rejected offers



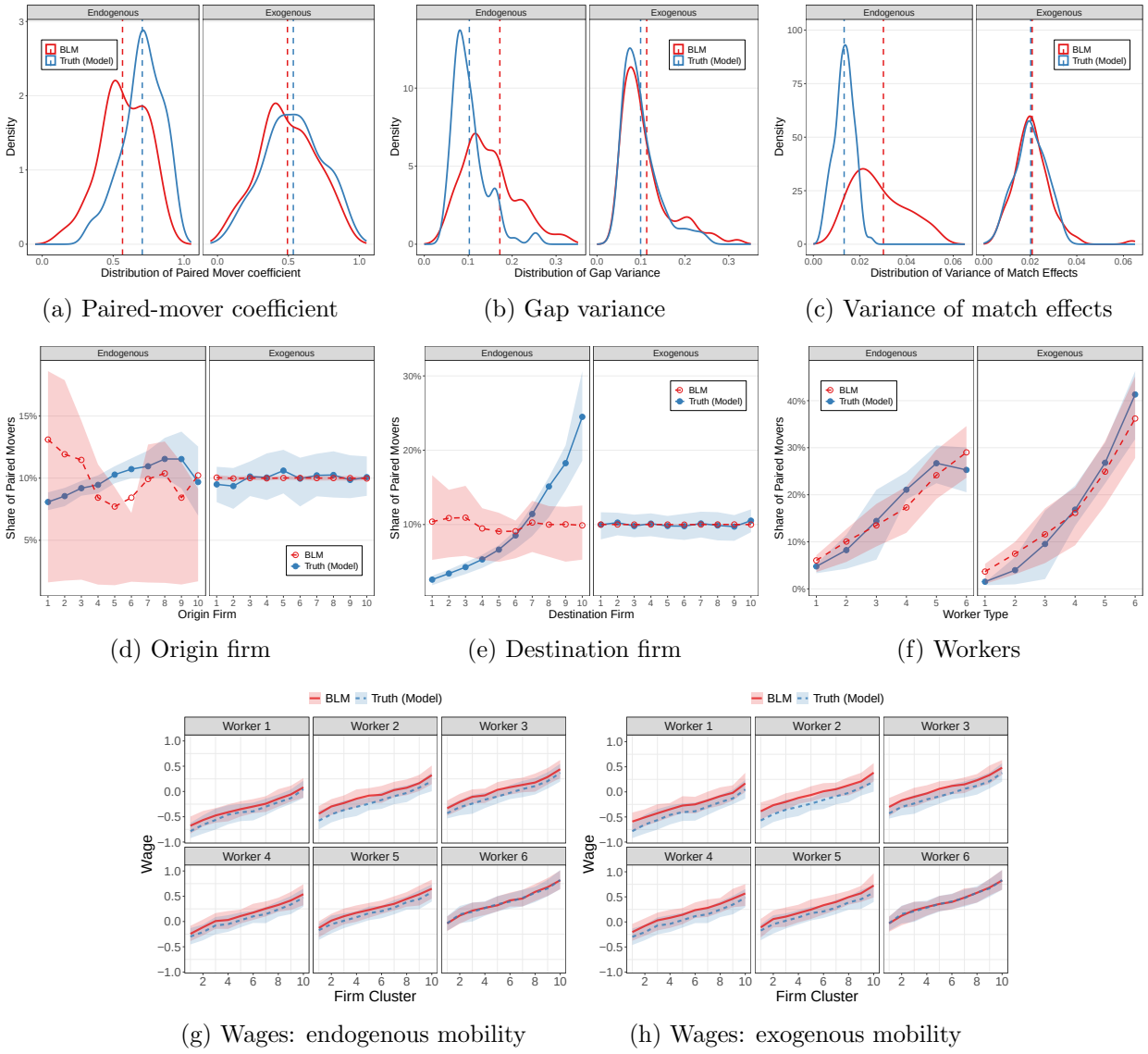
(b) Event study, 43% EE movers ($a = 0.13$)



(c) Event study, 97% EE movers ($a = 1.50$)

This figure shows results using data simulated from the alternative calibration in Table A10. In Panel A, we compare the change in firm effects (x-axis) to the change in wages (y-axis) for rejected offers. Panels B and C show the event study test in different samples. We plot the change in wages between destination and origin firms against the change in firm effects. Panel B uses the baseline on-the-job arrival rate parameter $a = 0.13$, while Panel C sets $a = 1.50$.

Figure A2: Monte Carlo simulations: BLM with discrete heterogeneity



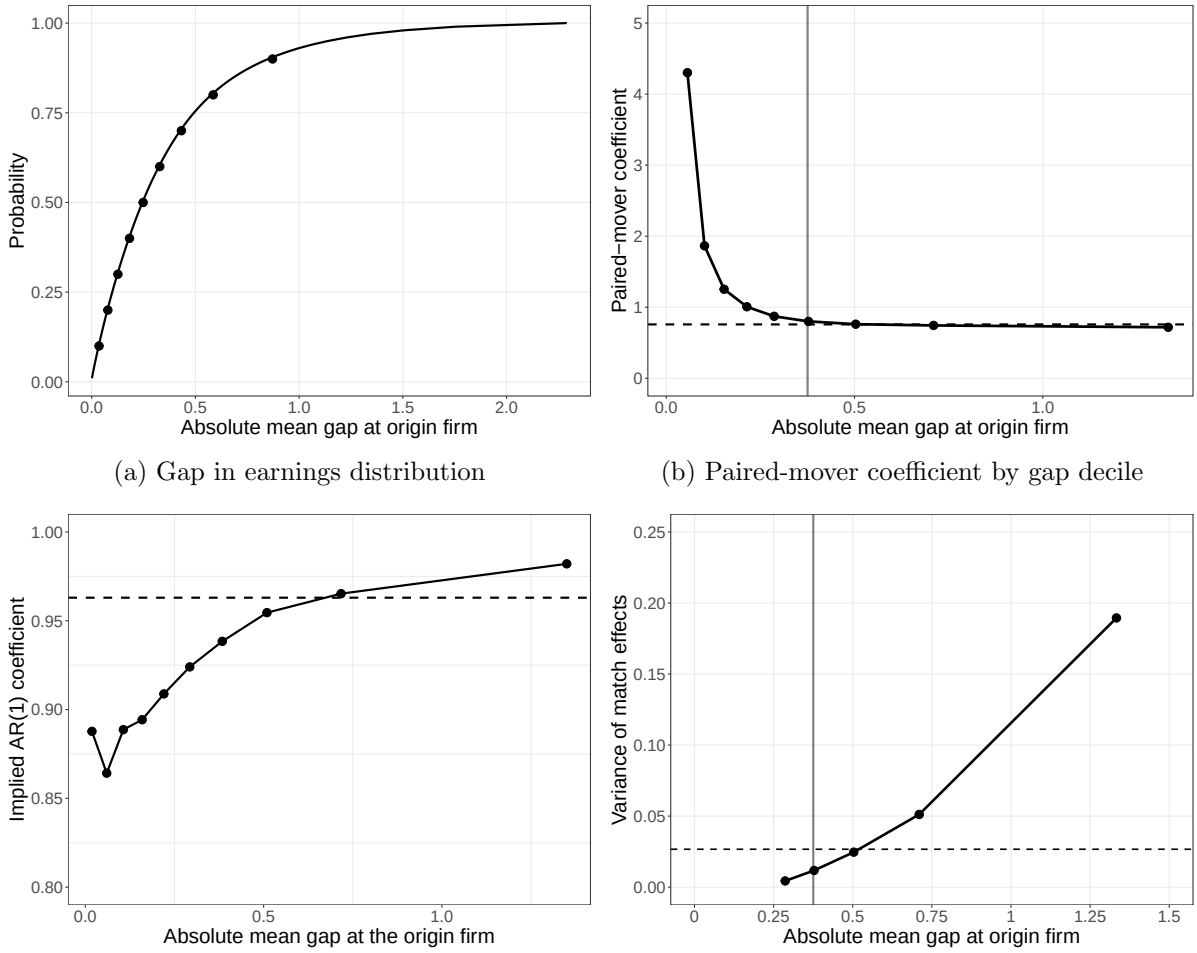
(a) Paired-mover coefficient (b) Gap variance (c) Variance of match effects

(d) Origin firm (e) Destination firm (f) Workers

(g) Wages: endogenous mobility (h) Wages: exogenous mobility

This figure shows the distribution of the paired-mover coefficient across BLM estimations. We draw 100 simulations each of the endogenous and exogenous versions of the model with discrete heterogeneity (6 worker types and 10 firm types) and estimate the BLM model in each simulation. In Panels A–C, we plot the distribution of the paired-mover coefficient (Panel A), the gap variance (Panel B), and the variance of match effects implied by the paired-mover coefficient under exogenous mobility (Panel C) in the true data (fed into BLM) and from data simulated from the estimated BLM model. In each panel, the dashed vertical line shows the average value in the true and estimated data across simulations. In Panels D–F, we plot the average distribution of origin firms (Panel D), destination firms (Panel E), and workers (Panel F) across simulations. Panels G and H show the average wage for each worker type and firm type under endogenous and exogenous mobility, respectively, in the true data (simulated from our model) and in the estimated BLM model. In each panel, bands represent p25 and p75 across simulations.

Figure A3: Paired movers across the gap distribution



(a) Gap in earnings distribution

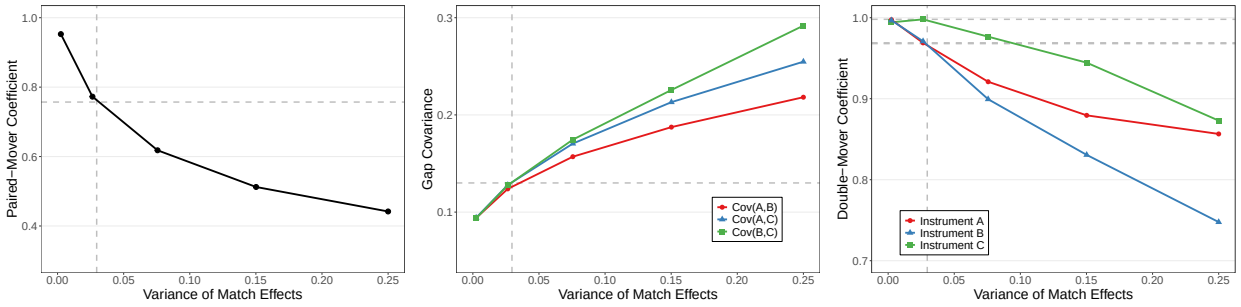
(b) Paired-mover coefficient by gap decile

(c) Implied AR(1) by gap decile

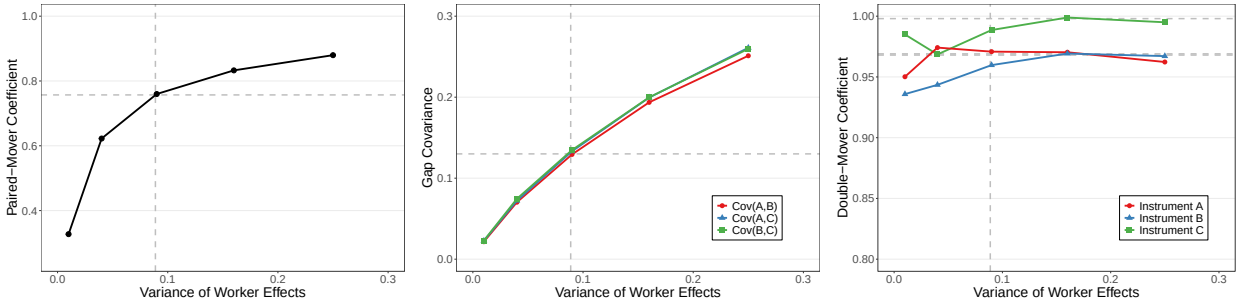
(d) Variance of match effects

This figure shows paired-mover statistics across the distribution of the absolute gap in earnings at the origin firm. We define the deciles based on the lagged earnings gap. Panel A shows the cumulative distribution of the absolute gap in earnings in the last year at the origin, where the dots mark the cutoff of each decile. Panel B shows the paired-mover coefficient in each decile of the gap distribution. The first decile is excluded due to large (imprecise) estimates. The dashed horizontal line shows the baseline estimate from Column 1 of Table 4, and the vertical line shows the average absolute gap. Panel C shows the implied AR(1) coefficient of the log earnings gap at the origin firm. The dashed horizontal line shows the AR(1) coefficient estimated using the full sample (Appendix Table A4). Panel D shows the variance of match effects under the assumption of exogenous mobility according to Equation 3. The first five deciles are not shown since the implied variance is negative. The dashed horizontal line shows the variance of match effects implied by the model (Table 7). In Panel D, summing up the variances across the five deciles and dividing by ten yields a mean implied variance of 0.028. Appendix Table A16 shows the numbers underlying Panels B and D.

Figure A4: Paired- and double-mover moments and model parameters



(a) Paired-mover coefficient *vs.* σ_m^2 (b) Gap covariance *vs.* σ_m^2 (c) Double-mover coefficient *vs.* σ_m^2



(d) Paired-mover coefficient *vs.* σ_h^2 (e) Gap covariance *vs.* σ_h^2 (f) Double-mover coefficient *vs.* σ_h^2

This figure shows heuristic identification plots. In each panel we start with the estimated parameter (shown in the vertical dashed line). We then vary that estimated parameter (holding all other model parameters constant) and show how changing that parameter affects model moments. We focus on two key parameters and three key moments. Panels A–C show how the variance of match effects (σ_m^2) affects the paired-mover coefficient (Panel A), the covariance of gaps across the three firms of paired double movers (Panel B), and the double-mover coefficients (Panel C). Panels D–F show how the variance of person effects (σ_h^2) affects the same three moments.